

Review

The central role of colloids to explain the crystallization dynamics of halide perovskites: A critical review

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SUMMARY

Solar photovoltaics offers a promising avenue for sustainable power generation, with perovskite solar cells gaining significant interest due to their cost effectiveness and remarkable performance. Lead-based perovskites, while possessing superior electron mobility and carrier properties, raise concerns due to environmental and health risks. Substituting lead with tin holds promise. Tin halide perovskites, particularly with iodine as the halide, exhibit lower band gaps, suggesting heightened efficiency. However, managing the crystallization process, crucial for achieving films with desired morphological properties and crystal phase purity, poses a significant challenge for perovskite materials. Although solvent and additive engineering are extensively explored, their specific effects on the colloidal properties of perovskite suspensions received relatively little attention. This review aims to provide an overview of the world of colloids, shedding light on how the manipulation of colloidal properties in perovskite suspensions can exert a substantial influence on the crystallization kinetics of halide perovskite thin films.

INTRODUCTION

Solar photovoltaics (PV) has emerged as a highly promising renewable energy production technology, presenting significant potential for sustainable electricity generation. Within the PV sector, perovskite solar cells have gained substantial attention in recent years, emerging as a “rising star” owing to their manufacturing cost effectiveness and exceptional performance capabilities.^{1–3} Among the metal halide perovskites (MHPs), lead-based perovskites have garnered extensive research focus due to their remarkable electron mobility, long charge carrier diffusion length, and extended carrier lifetime.^{4,5} Nevertheless, the toxicity associated with lead poses environmental and health risks.⁶ To address this concern, substituting lead with an alternative element is a plausible solution. Tin, belonging to the same periodic table group, stands as the most viable replacement option. Tin halide perovskites (THPs) exhibit a lower band gap compared to their lead counterpart, suggesting possible higher efficiency in accordance with the Shockley-Queisser theory.⁷ Furthermore, their low band gap makes them particularly attractive for use in tandem solar cells.⁸ However, challenges persist in attaining precise control over the crystallization dynamics of THPs due to their high crystallization rate, affecting their efficiency.⁹ Nowadays, THPs achieve a maximum efficiency of approximately 15%, motivating ongoing research efforts aimed at enhancing their performance.¹⁰

PROGRESS AND POTENTIAL

Recently, there has been a growing interest in seeing the perovskite precursor solution as a colloidal suspension rather than a chemical solution. However, little attention has been paid to how these colloids interact to form perovskite. This review aims to shed light on this aspect, focusing mainly on the effect of additives and solvents on the properties of colloidal suspensions. Colloid theory emerges as a tool for comprehending perovskite crystallization mechanisms, thereby advancing the development of improved perovskite-based solar cells.

Among the array of methodologies investigated to finely tune the crystallization kinetics and morphology of MHP films, solvent engineering and additive incorporation have emerged as highly influential techniques within the field of solution engineering.¹¹ By solvent engineering, we refer to the selection of solvents, which, based on their physical and chemical properties, are able to interact with perovskite salts.^{12–19} Conversely, additive incorporation involves the introduction of additives, usually in low concentrations, that exhibit pronounced capabilities in modulating the growth dynamics of a perovskite solid film.^{10,20–23} Since the crystalline thin film originates directly from the suspension, acquiring a meticulous understanding of the underlying crystallization mechanism is crucial to obtain effective control over perovskite formation. In this pursuit, colloid theory has proven to be an invaluable asset. Recent years have witnessed an increasing impetus to perceive the perovskite solution as a colloidal suspension rather than a mere chemical solution, gaining traction as a possible path for furthering the science.^{24–27} We are aware that the principles expounded in this analysis have broad applicability and can be generalized to a variety of sectors that use perovskites as thin films, including light-emitting diodes and transistors.^{28,29} Nonetheless, our review only focuses on the application of MHPs in the solar cells field.

This comprehensive review seeks to establish a pivotal connection between the properties of colloidal suspensions and the intricate crystallization dynamics of perovskite materials. The review unfolds across four sections: (1) a quick introduction to fundamental colloidal theory and an explanation of the principles regulating colloidal suspension stability, with emphasis on the Derjaguin-Landau-Verwey-Overbeek (DLVO) theory as a framework; (2) an extensive examination of various techniques dedicated to the characterization of colloidal suspensions; (3) a concise overview of pioneering investigations that analyzed and conceptualized the structure of the perovskite colloid; (4) a discussion of the profound influence of additives and solvents on perovskite suspensions in which we analyze their roles in modulating the crystallization processes; and (5) a brief introduction of the key simulation techniques tailored for the study of colloidal suspensions.

OVERVIEW OF COLLOID THEORY

The term colloid finds its origin in the Greek words “kolla” and “oid,” signifying “glue” and “like,” respectively.³⁰ A colloidal dispersion entails two distinct phases: the dispersed phase, also known as the internal phase, and the external phase, defined as the dispersion medium. Colloidal particles (dispersed phase) typically fall within the size range of 1 nm to 1 μ m. These particles experience the influence of the thermal fluctuations in the surrounding medium, a phenomenon referred to as “Brownian motion,” which prevents their rapid settling or buoyancy. Colloidal systems find extensive relevance in various fields, including environmental applications, mining and mineral processing,^{31,32} manufacturing and materials science applications,^{33–35} and food and personal care products,^{36–38} as well as biological and medical applications.^{39,40} Numerous types of colloidal systems exist, depending on the nature of the internal and external phases.^{41–44} The most common types include solid aerosols (dispersion of solid particles in a gas), liquid aerosols (dispersion of liquid droplets in a gas), foams (dispersion of gas bubbles in a liquid), sols (dispersion of solid particles in a liquid), emulsions (dispersion of liquid droplets in another liquid), and gels (dispersion of a liquid in a solid matrix).

In the study of colloidal suspension dynamics, the interaction potential between two particles in close proximity is crucial.^{45,46} From a thermodynamic perspective,

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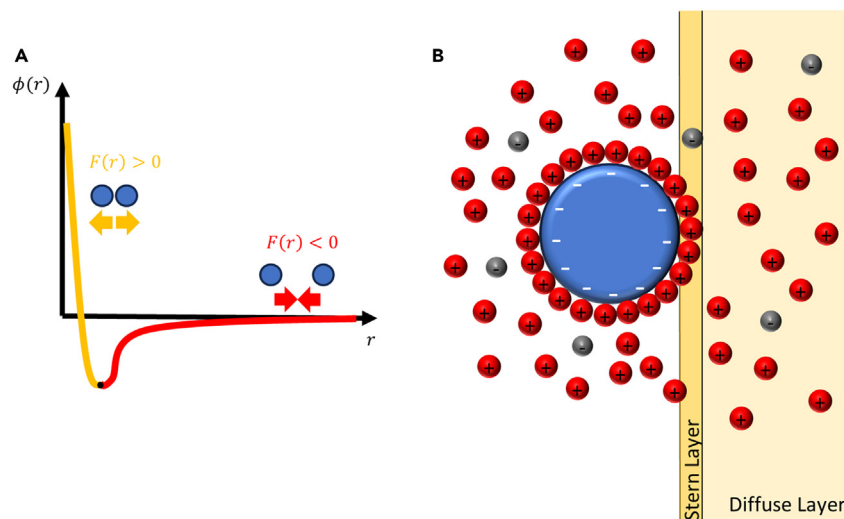


Figure 1. Example of the interaction potential between colloids and the EDL

(A) Interaction potential between colloids: the yellow portion represents the repulsive force (negative slope), whereas the red portion represents the attractive force (positive slope).
(B) A colloidal particle surrounded by charges forming the electrical double layer (EDL).

nanometer-sized colloidal particles commonly exhibit a natural tendency to aggregate due to attractive forces, leading to the formation of clusters or larger agglomerates. To maintain colloidal suspensions in a non-aggregated state, repulsive interparticle forces must be carefully considered. When the repulsive forces prevail over the attractive ones, the colloidal suspension is called “kinetically stable.” The main interactions governing colloidal systems can be categorized into hydrodynamic interactions, dispersion forces (attractive), surface forces (repulsive), and depletion forces.^{47–49} For the aim of this review, we will consider only the dispersive and surface forces.

Before going into a detailed discussion of the forces, let us consider a general scenario involving the interaction potential $\phi(r)$ between two colloidal particles, where r is the distance between their centers. For simplicity, we consider one colloid fixed at the origin of a spherical system of coordinates, whereas the other is free to move along the radial coordinate. An illustration of an interaction potential is presented in Figure 1A, and the force $F(r)$ acting between the two colloidal particles can be expressed as⁵⁰

$$F(r) = - \frac{\partial \phi(r)}{\partial r}. \quad (\text{Equation 1})$$

This potential profile can be divided into two distinct regions: a decreasing portion and an increasing portion, corresponding to a repulsive and an attractive force regime, respectively. The minimum point, in which the derivative of $\phi(r)$ is zero, represents a stable equilibrium distance, implying that this configuration results in flocculation within the colloidal suspension. Flocculation is a scenario in which colloids do not come into contact (as in coagulation) but instead form agglomerates, known as “flakes.” Flocculation is a reversible process, whereas coagulation is irreversible.

As previously mentioned, attractive forces include “induced dipole-induced dipole” interactions, more commonly referred to as “London dispersion forces” (van der Waals forces).⁵¹ The magnitude of these forces depends on several variables, including the separation distance between colloidal particles, the polarizability (which denotes the capacity of a molecule to induce distortions in its electronic

cloud, forming a dipole moment), and the ionization potential (a measure of the energy required to remove the most loosely bound electron from an atom or molecule). The attractive potential between two molecules at a distance d_{12} can be mathematically expressed as follows:

$$\varphi(d_{12}) = -\frac{\mathcal{A}_{12}}{d_{12}^6}, \quad (\text{Equation 2})$$

where $\mathcal{A}_{12}$ is the interaction. Van der Waals forces between two spherical colloidal particles characterized by radii R_1 and R_2 can be computed by summing the contributions originating from all the molecules constituting the particles. The final expression for the attractive potential is

$$\Phi_a(D) = -\frac{\pi^2 \rho_1 \rho_2}{6} C \left[\frac{2R_1 R_2}{c^2 - (R_1 + R_2)^2} + \frac{2R_1 R_2}{c^2 - (R_1 - R_2)^2} + \ln \left(\frac{c^2 - (R_1 + R_2)^2}{c^2 - (R_1 - R_2)^2} \right) \right], \quad (\text{Equation 3})$$

with ρ_1 and ρ_2 representing the electron densities of the particles, c the center-to-center distance between the two particles, and $D = c - (R_1 + R_2)$ the distance between the two surfaces.⁴⁸ If the two particles have the same radius R and are very close to each other ($D \ll R$), then the expression can be simplified as follows:

$$\Phi_a(D) = -\frac{\pi^2 \rho_1 \rho_2 C R}{12D} = -\frac{AR}{12D}, \quad (\text{Equation 4})$$

where A is the Hamaker constant.

As elucidated earlier, the presence of repulsive forces between colloids is imperative to avoid particle aggregation. Among these repulsive forces, it is crucial to mention electrostatic forces originating from the electrical double layer (EDL). In practice, the surface of colloidal particles is typically not electrically neutral but rather carries a net charge that can be attributed to the ionization of surface groups or the adsorption of ions from the surrounding solution. To maintain the overall electrical neutrality of a colloidal suspension, the dispersing medium must provide an equivalent number of charges with opposite polarity, enveloping the colloidal particle's surface. The layer composed of opposite charges in the surrounding liquid is denoted as the EDL.^{52,53} As depicted in Figure 1B, the EDL consists of two distinct regions: the innermost portion comprises ions or charges that are strongly bound to the colloid's surface, commonly known as the Helmholtz or Stern layer, whereas the region adjacent to this layer encloses charges that are weakly bound to the colloid, forming the diffuse layer. The thickness of the double layer is quantifiable and is referred to as the Debye length (ℓ), which can be determined through electrostatic principles.^{54,55} For simplicity, let us consider a planar plate. The electric potential $\psi(x)$ at a distance x from the plate can be expressed, in accordance with the Poisson equation, as follows:

$$\epsilon_r \epsilon_0 \frac{d^2 \psi(x)}{dx^2} = -\rho(x), \quad (\text{Equation 5})$$

where ϵ_r is the relative permittivity of the dispersing medium, ϵ_0 is the dielectric constant of vacuum, and ρ is the charge density. If we assume that the distribution of electrolytes adheres to the Boltzmann's law and that the electric potential ψ is of low magnitude (a condition often referred to as "Debye-Hückel regime"), we can utilize the linearized version of the Poisson-Boltzmann equation:

$$\frac{d^2 \psi(x)}{dx^2} = \kappa^2 \psi(x). \quad (\text{Equation 6})$$

The Debye-Hückel constant κ appearing in Equation 6 can be written as:

$$\kappa = \left[\frac{1}{\epsilon_r \epsilon_0 k_B T} \sum_i z_i^2 e^2 N_{i\infty} \right]^{1/2}, \quad (\text{Equation 7})$$

where k_B is the Boltzmann constant, T is the absolute temperature, z_i is the valency of the ion (or counter ion) i , e is the elementary charge, and $N_{i\infty}$ is the density number of the ion i at $x \rightarrow \infty$ (i.e., in the bulk of the solution). The Debye screening length, which measures the thickness of the diffuse layer, is defined as the reciprocal of the Debye-Hückel constant, so $l = \kappa^{-1}$. The simple solution for Equation 6 is

$$\psi(x) = \psi_0 \exp(-\kappa x), \quad (\text{Equation 8})$$

where ψ_0 is the surface potential at $x = 0$. From Equation 8, it is evident that as the concentration of ions in the solution rises, the thickness of the double layer reduces; thus, the surface potential decreases faster. By superimposing the effects of two EDLs, the interaction potential attributed to electrostatic forces can be derived. In the case of two spherical particles with radius R situated at a separation distance D , the repulsive potential can be expressed as

$$\Phi_r(D) = \frac{64N_0k_B T \pi R \gamma^2}{\kappa^2} \exp(-\kappa D), \quad (\text{Equation 9})$$

where N_0 is the ion density number at $x = 0$ and $\gamma = \tanh(ze\psi_0/4k_B T)$ is the reduced surface potential.

In the 1940s, Derjaguin, Landau, Verwey, and Overbeek pioneered the development of the DLVO theory, a seminal model in the field of colloid physics and chemistry.⁴² The fundamental hypothesis of this theory is that it is possible to sum attractive and repulsive potentials to formulate a unified potential that comprehensively characterizes the dynamics of colloidal systems. Therefore, if we assume two equal and very close spherical particles, we can combine the two potentials and obtain a total interaction potential as follows:

$$\Phi_{\text{total}} = \Phi_a + \Phi_r = -\frac{AR}{12D} + \frac{64N_0k_B T \pi R \gamma^2}{\kappa^2} \exp(-\kappa D). \quad (\text{Equation 10})$$

Upon scrutinizing this expression as a function of interparticle separation, it becomes evident that there is a well-defined maximum point, which represents the aggregation barrier (see Figure 2A). When the thermal energy associated with colloidal particles (Brownian force, $k_B T$) exceeds the energy barrier, the colloids aggregate by positioning themselves at a distance smaller than the barrier (going toward the aggregation). Conversely, if the energy is insufficient to overcome this barrier, then the colloids will persist at a distance larger than the position of the barrier, where repulsive forces predominate over attractive forces.

The total interaction potential is primarily influenced by two key factors: ion concentration within the suspension and particle size.^{48,49} Ion concentration exclusively influences electrostatic repulsion, such that an increase of it, achieved, for instance, by introducing a salt into the suspension, diminishes the EDL, consequently reducing electrostatic repulsions and promoting colloidal system destabilization.⁵⁶ There exists a critical salt concentration, known as the critical coagulation concentration, beyond which no aggregation barrier persists. As depicted in Figure 2B, a reduction in the concentration of salts in the liquid phase correlates with an elevation in the energy barrier to colloidal aggregation. Regarding colloid size, both the potentials depend on colloid dimensions, with a more pronounced influence on electrostatic repulsion. Consequently, it can be generally stated that larger colloids exhibit higher aggregation barriers (Figure 2B).

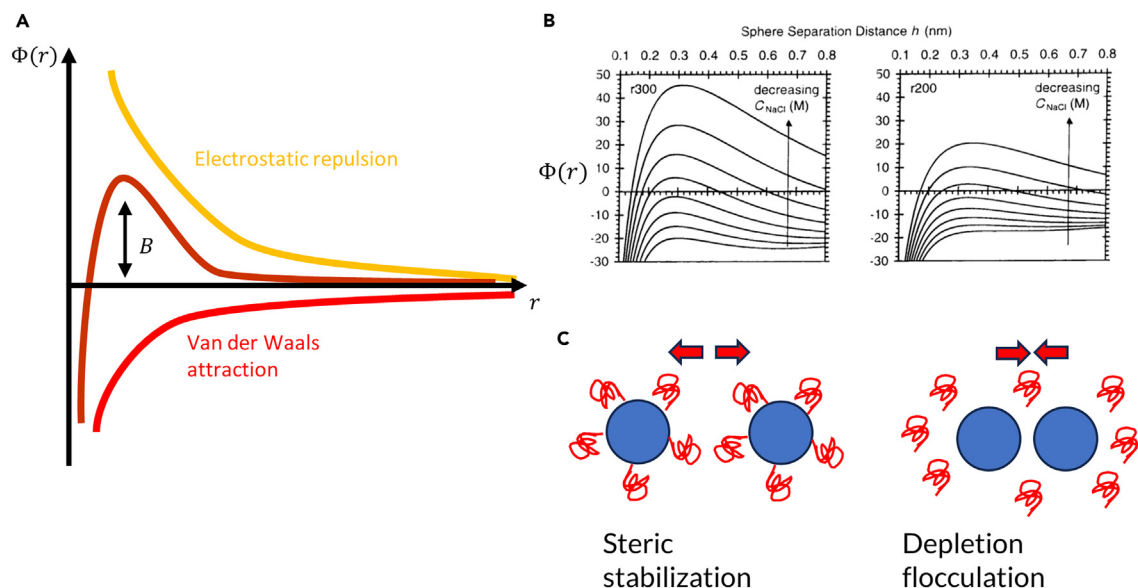


Figure 2. Colloidal interaction potential and effects of salts and polymers

(A) Total interaction potential (dark red curve) obtained as the sum of electrostatic repulsive potential (yellow curve) and van der Waals attractive potential (red curve). B represents the aggregation barrier.

(B) Effect of NaCl concentration on an aqueous solution of polymer latices. The difference between the two diagrams is the size of the colloidal particles (radius of 300 nm on the left and 200 nm on the right) (adapted with permission from Hanus et al.⁵⁶ Copyright 2001, American Chemical Society).

(C) Steric stabilization and depletion flocculation induced by polymers.

Another mechanism to prevent colloidal system destabilization, which falls beyond the domain of the DLVO theory, pertains to stabilization via steric repulsion (Figure 2C). In practice, this entails the adsorption of large molecules, such as polymers or surfactants, onto colloidal particles, effectively forming a physical protective layer.⁵⁷ It is noteworthy that these repulsive effects are applicable when polymers adhere to colloids. Conversely, when polymers are freely suspended in the dispersion, the opposite effect could occur due to osmotic pressure, favoring colloidal particle aggregation.⁵⁸ In the presence of steric repulsion, the total interaction can be formulated as $\Phi_{\text{total}} = \Phi_a + \Phi_r + \Phi_s$, where Φ_s is the steric potential and acts in the same directions as Φ_r .

CHARACTERIZATION OF COLLOIDAL SUSPENSIONS

Enhancing our understanding of crystallization mechanisms necessitates an investigation into the properties of perovskite colloidal suspensions. Nowadays, numerous techniques are available to study the behavior of colloidal suspensions. These include dynamic light scattering (DLS), zeta potential analysis, small-angle neutron scattering (SANS), and small-angle X-ray scattering (SAXS). These techniques provide valuable insights into the physical and chemical characteristics of colloidal suspensions.

DLS is a physical technique used for the determination of the size distribution of colloidal particles in a suspension.^{59,60} This method exploits two fundamental characteristics of colloids: the Tyndall effect, which manifests as light scattering, and the Brownian motion exhibited by the particles. When a colloidal particle is illuminated by a light beam, it scatters light, functioning as a secondary light source. The scattered light is collected by a detector, as shown in Figure 3A. Given the Brownian motion of particles, the distance traveled by the scattered light from the particle to the detector varies with time. Consequently, the net intensity exhibits random fluctuations over time. In DLS, the characteristic time of these intensity fluctuations, which

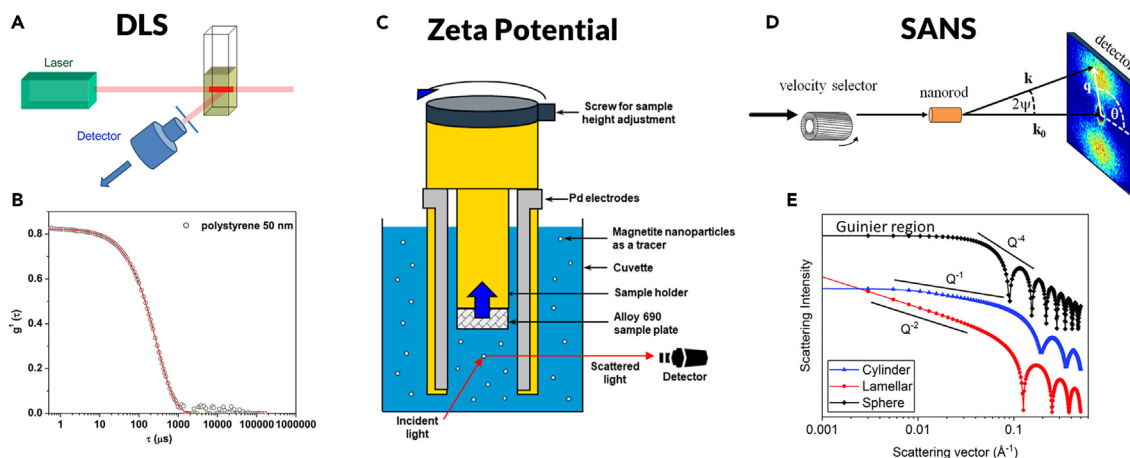


Figure 3. Characterization techniques for colloidal suspensions

(A) Dynamic light scattering setup (adapted with permission from Hassan et al.⁵⁹ Copyright 2015, American Chemical Society).

(B) Autocorrelation function of the light scattered by a dispersion containing polystyrene particles with a size of 54 nm. The red curve is the exponential decay fit (reprinted with permission from Hassan et al.⁵⁹ Copyright 2015, American Chemical Society).

(C) Scheme of zeta potential measurement kit for the study of the electrokinetic properties of magnetite particles (reprinted from Lee et al.,⁶¹ license link: <https://creativecommons.org/licenses/by/4.0/>).

(D) Small-angle neutron scattering setup (adapted from Vivas et al.⁶²).

(E) Calculated scattering patterns for objects of different shapes (reproduced from McDowall et al.⁶³).

is directly linked to the diffusion coefficient of the colloids, is measured. The smaller the particle, the more rapid the intensity fluctuations. On the other hand, larger particles diffuse more slowly. If we represent the scattered intensity at an arbitrary time as $I(t)$ and that after a delay time τ as $I(t + \tau)$, we can write the normalized autocorrelation function as

$$g^2(\tau) = \frac{\langle I(t)I(t+\tau) \rangle}{\langle I(t) \rangle^2}, \quad (\text{Equation 11})$$

where $\langle f(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(t) dt$. When $\tau = 0$, the signal is perfectly correlated, and

the numerator is trivially equal to $\langle I^2 \rangle$; when $\tau \rightarrow \infty$, the signal is completely uncorrelated, and the numerator is equal to $\langle I \rangle^2$, as $\langle I(t)I(t + \tau) \rangle = \langle I(t) \rangle \langle I(\tau) \rangle$. For photo counts obeying Gaussian statistics, $g^2(\tau)$ is related to the first-order correlation function of the electric field $g^1(\tau)$ by the relationship $g^1(\tau) = [g^2(\tau) - 1]^{0.5}$. If we assume that the suspension is made of monodisperse spherical particles undergoing Brownian motion, then the autocorrelation function decays exponentially with the delay time τ as follows:

$$g^1(\tau) = A \cdot e^{-Dq^2\tau} + A_0, \quad (\text{Equation 12})$$

where A is the amplitude of the correlation function, A_0 is the baseline, D is the translational diffusion coefficient, and $q = (4\pi n / \lambda) \sin(\theta / 2)$ is the magnitude of the scattering vector, with n the refractive index of the medium, λ the wavelength of the light, and θ the scattering angle. Assuming that the spherical particles are larger than the solvent molecules and that the interparticle interactions are negligible, the hydrodynamic radius R_h can be obtained from the diffusion coefficient through the well-known Stokes-Einstein relationship:

$$D = \frac{k_B T}{6\pi\eta R_h}, \quad (\text{Equation 13})$$

where η is the solvent viscosity. Through the analysis of the autocorrelation function, it becomes feasible to determine the diffusion coefficient of colloidal particles using fitting procedures, thereby providing insights into the particle size (Figure 3B). As previously stated, the underlying assumption is that the colloidal suspension is only formed by non-interacting spherical particles of the same size. However, in many real-world systems, colloidal suspensions exhibit a particle size distribution. Consequently, the right-hand side of Equation 12 becomes a weighted sum of exponentials, accounting for the different amplitudes associated with different particle sizes. In cases of monomodal distributions, the cumulant method is typically employed to analyze the data and extract relevant information.⁶⁴ On the other hand, the non-negative least-squares method is utilized for systems with multimodal distributions.⁶⁵ DLS offers several advantages, including high accuracy, repeatability, the ability to measure very small volumes, and the capability to analyze turbid samples. However, a notable drawback of this technique is the limited ability to directly measure the true size of particles, as DLS provides information primarily about the particles' hydrodynamic radius, which is dependent on the interaction between the particles and the surrounding solvent.

The zeta potential, also referred to as the “electrokinetic potential,” represents the potential difference between the bulk fluid surrounding the dispersed particle and the fluid layer containing the oppositely charged ions that is closely associated with the nanoparticle surface.⁶⁶ This parameter serves as an indicator of the colloidal suspension's stability, with higher zeta potential values indicating stronger electrostatic repulsion forces between particles. The measurement of zeta potential can be conveniently performed in conjunction with DLS measurements using a technique known as electrophoretic light scattering (ELS). By applying an electric field, particles exhibit electrophoretic mobility, causing them to migrate toward an electrode bearing an opposite charge, as depicted in Figure 3C.^{48,61,67,68} It can be shown that the particles move with a terminal velocity $v = \mu E$, where μ is the electrophoretic mobility and E is the electric field. When a light of frequency ν and wavelength λ is incident on a moving particle in an applied electric field, the scattered light will undergo a Doppler frequency shift $\Delta\nu$, which can be written as follows:

$$\Delta\nu = \frac{q \cdot v}{2\pi} = \frac{4\pi n \sin(\theta) \cos(\theta) v}{2\pi \lambda} = \frac{n \sin(2\theta) v}{\lambda}. \quad (\text{Equation 14})$$

From this equation, we can write electrophoretic mobility as:

$$\mu = \frac{v}{E} = \frac{\lambda \Delta\nu}{n E \sin(2\theta)}. \quad (\text{Equation 15})$$

With an accurate measurement of the frequency shift, it is possible to use Equation 15 to derive the electrophoretic mobility, which, in turn, is related to the zeta potential by the well-known Henry equation, reading

$$\zeta = \frac{3\mu\eta}{2\epsilon_0\epsilon_r f(\kappa R)}, \quad (\text{Equation 16})$$

where $f(\kappa R)$ is the Henry function, which depends on the inverse of the Debye length κ^{-1} and the radius of the particle R . Nowadays, most instruments are based on the phase analysis light scattering technique, which measures phase shift rather than frequency shift.⁶⁹

SANS and SAXS are powerful techniques employed for investigating the structural characteristics of materials within the mesoscopic scale, spanning length scales ranging from 10 to 1,000 Å.^{70–72} In a SANS experiment, a beam of neutrons is

focused onto the sample. Due to the interactions with the atomic nuclei and the magnetic moment of unpaired electrons, the neutrons are elastically scattered,⁶² as depicted in Figure 3D. Conversely, in a SAXS experiment, X-rays are scattered by the electrons within the atoms constituting the sample. The intensity of the scattered radiation measured by the detector arises from the interference of multiple scattered waves as a function of the scattering angle θ . The scattered intensity can be expressed as a function of the scattering vector q as follows:

$$I(q) = N_d V^2 (\rho_p - \rho_s)^2 P(q) S(q), \quad (\text{Equation 17})$$

where N_d is the number density of particles, V is the volume of a particle, ρ is the scattering length density (the subscript “p” and “s” referring to particle and solvent, respectively), $P(q)$ is the intraparticle structure factor, and $S(q)$ is the interparticle structure factor. The scattering length density characterizes the strength of the interaction between the nuclei comprising a particle and the incident neutrons. The intraparticle structure factor $P(q)$ encodes information about the size and shape of the particles, whereas the $S(q)$ factor accounts for interparticle interactions. The specific form of the interparticle function $S(q)$ plays a crucial role in determining the overall scattering intensity. This influence manifests as changes in the slope of $I(q)$ at low q values and the appearance of interaction peaks. In the literature, various mathematical expressions for $P(q)$ and $S(q)$ can be found for different geometric shapes and diverse types of interactions.⁷³ When $P(q)$ is plotted on a double logarithmic scale, it exhibits two distinct regimes: a plateau at low q followed by a decrease at higher q values. Notably, it has been demonstrated that, at small q values, the intensity obeys a universal law, called Guinier’s approximation, that is independent of the specific particle shape, reading^{74–76}

$$\ln[I(q)] \propto \frac{q^2 R_g^2}{3}, \quad (\text{Equation 18})$$

where R_g is the radius of gyration of the particles. Guinier’s approximation establishes a universal and model-independent method for determining particle sizes based on small-angle scattering data. By analyzing the decreasing trend of the intraparticle factor beyond the range of validity of Guinier’s law, qualitative information about the particle structure can be obtained. In this region, the intensity typically follows an exponential decay of the form $I \approx q^{-\alpha}$. Figure 3E portrays distinct profiles of scattering intensity corresponding to different exponents α , associated with different geometries. For instance, an exponent 1 indicates a structure composed of randomly oriented elongated objects, an exponent 2 suggests a structure consisting of randomly oriented flat objects, and an exponent of 4 represents a system made of spherical objects. SANS offers several notable advantages, including its ability to provide nanoscale information, its non-destructive nature, and its capacity to perform experiments with contrast variation by utilizing isotopic labeling for different components. However, there are certain limitations associated with SANS. Firstly, it requires access to nuclear research facilities, whose availability can be limited. Secondly, the neutron flux used in SANS experiments is relatively low, resulting in longer test durations and making it challenging to study time-dependent processes. While being closely related scattering techniques, SANS and SAXS offer distinct properties and provide complementary information. The choice between them depends on the specific problem under investigation. Both offer nanoscale insights and non-destructive measurements. SAXS boasts a high photon flux allowing for significant spatial and temporal resolutions. However, it faces limitations, including restricted access to synchrotron facilities, a shallower penetration depth, and more intricate data analysis.

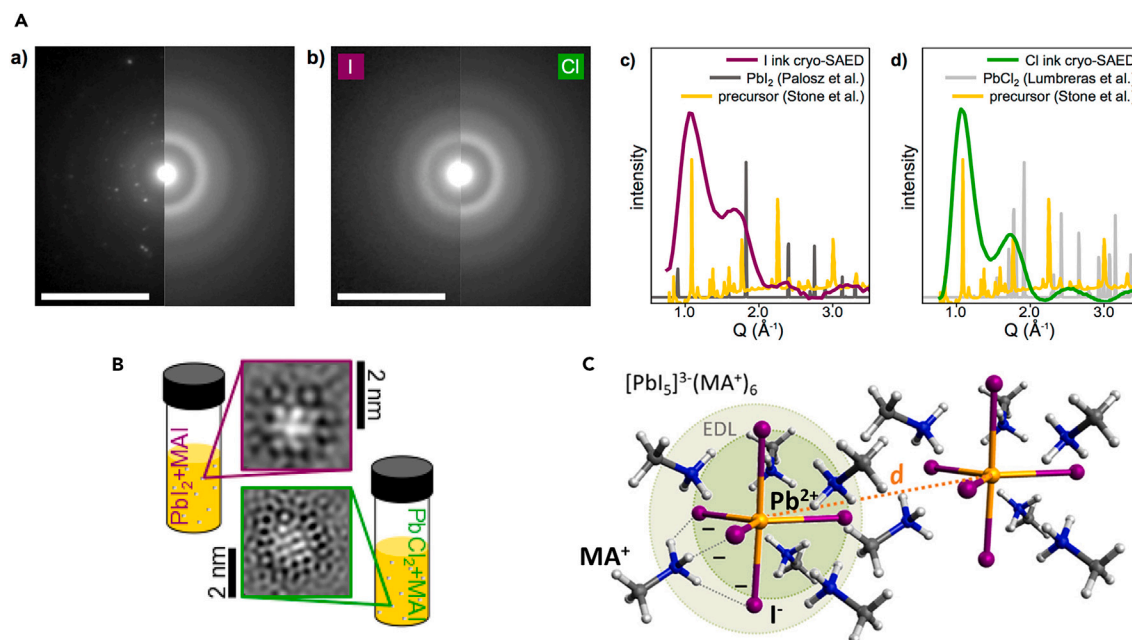


Figure 4. Perovskite colloid

(A) Cryo-selected area electron diffraction analysis for MAPbI₃ (a and left side of b) and MAPbCl₃ (right side of b) inks. Radially integrated patterns of MAPbI₃ (purple curve) (a) and MAPbCl₃ (green curve) (b) inks. The gray diffraction patterns are for PbI₂ (c) and PbCl₂ (d) salts. The yellow diffraction patterns are those of MA₂PbX₄ crystalline structures obtained by Stone et al.⁷⁹ (adapted with permission from Dutta et al.⁷⁷ Copyright 2020, American Chemical Society).

(B) Cryo-EM single-particle analysis images of perovskite colloids crystal structures (adapted with permission from Dutta et al.⁷⁷ Copyright 2020, American Chemical Society).

(C) Scheme of the perovskite colloid and the EDL around it (adapted with permission from Flatken et al.⁸⁰ Copyright 2022, American Chemical Society).

EXPLORING THE NATURE OF PEROVSKITE COLLOIDS

At this point, understanding the nature of perovskite colloid becomes critical. Literature on this topic is relatively recent, with the pioneering investigation by Dutta et al. in 2020 representing the initial attempt to characterize the colloid's nature.⁷⁷ Other attempts have been made in the past utilizing scattering and diffraction techniques; however, these were unable to go into great depth regarding the structure of the perovskite colloid.⁷⁸ In their study, Dutta et al. employed cryoelectron microscopy (cryo-EM) techniques to analyze the perovskite colloid. Two distinct suspensions, MAPbI₃ and MAPbCl₃, in DMF were meticulously studied. Figure 4A displays the results of cryo-selected area electron diffraction conducted on both samples (Figure 4A, panels a and b). Notably, both samples exhibit discernible crystal structures, regardless of the ink type. By radially integrating the spectra, it becomes evident that the diffraction data deviate from the patterns of the undissolved salts PbI₂ and PbCl₂ in suspension (Figures 4A, panels c and d), aligning more closely with the crystal structure of MA₂PbX₄ discovered by Stone et al. in 2018.⁷⁹ This result demonstrates that the perovskite colloid is neither amorphous nor undissolved precursor salts, but rather it is a crystalline material made up of the precursor salts. Cryo-EM single-particle analysis was used to unveil a plausible colloidal structure, as illustrated in Figure 4B, where brighter regions represent colloids and the dark background corresponds to the solvent. In a subsequent study, Flatken et al. used extended X-ray absorption fine structure (EXAFS) spectroscopy to analyze two suspensions: one containing PbI₂ in dimethylformamide/dimethyl sulfoxide (DMF/DMSO) and another comprising MAPbI₃ in the same solvent.⁸⁰ They determined distinct atomic environments for lead in PbI₂ compared to perovskite. Specifically,

in the PbI_2 case, lead predominantly coordinated with oxygen atoms from the solvent molecules, whereas in the perovskite case, lead was primarily surrounded by iodides. The observed chemical alteration indicates the transformative influence of organic cations on the system. To elucidate the specific role of organic cations, the authors conducted SANS analysis on both systems. This investigation demonstrated that the presence of organic cations in MAPbI_3 induces repulsions between colloids, evidenced by a peak in scattering intensity absent in the PbI_2 dispersion. Based on these findings, the authors delineated the structure of the perovskite colloid, as depicted in Figure 4C. The core of the colloid was identified as the negatively charged iodoplumbate ion, while the outer layer (EDL) consisted of organic cations, promoting repulsive interactions between colloids, along with solvent molecules.

ELUCIDATING PEROVSKITE FORMATION MECHANISMS VIA COLLOIDAL PROPERTIES

Within the perovskite field, achieving precise control over film morphology remains a hard challenge. Inadequate control results in films characterized by poor coverage and a heightened density of electronic traps, significantly reducing the solar cell's power conversion efficiency (PCE). In Figure 5, we report scanning electron microscopy (SEM) images taken from Yan et al.⁸¹ illustrating distinct films of PbI_2 (Figure 5A) and MAPbI_3 (Figure 5B) with different coordinating agents such as MAI and methylammonium chloride (MACI). With reference to PbI_2 films, a variation in the MAI: PbI_2 ratio results in a transition from needle-like crystals (characteristic of PbI_2 excess) to a more uniform morphology with reduced pinholes. With the addition of MACI, SEM analysis reveals perovskite films with significantly enhanced coverage and less pinholes compared to films without the additive at equivalent MAI: PbI_2 ratios. Device testing indicated a correlation between increased coverage and enhanced efficiency. For instance, the film with the 1:1 MAI: PbI_2 ratio exhibits an average (combined forward and reverse) efficiency of 7.71%, whereas the one incorporating MACI at 1:1:1 ratio demonstrates a doubled efficiency, reaching 15.01%. Munir et al. also investigated the impact of coordination chemistry on perovskite film morphology.⁸² Their findings, consistent with study previously mentioned, revealed that altering the ratio of MAI to PbI_2 or PbCl_2 influences the film morphology. In Figure 5C, *in situ* grazing-incidence wide-angle X-ray scattering (GIWAXS) during spin coating showed distinct outcomes: with a 1:1 MAI: PbI_2 ratio, GIWAXS peaks corresponding to the sol-gel phase and perovskite were observed around 21 s. Conversely, with a 3:1 MAI: PbCl_2 ratio, no discernible crystalline order was achieved during the rotation, and the system evolved dynamically over time. Subsequent SEM images after annealing revealed that, in the former case, film morphology remained unchanged with varying spin coating times beyond 21 s, while in the latter case, an incrementally enhanced degree of coating, as well as PCE, was observed.

In large-scale processes such as blade coating, precise control over morphology is essential. Zhong et al. investigated the ink-to-solid phase transformation MAPbI_3 films, elucidating the pivotal role of temperature.⁸³ Their findings demonstrate that temperature profoundly impacts the formation of perovskite solvate intermediates and consequently influences the final film morphology. Increasing temperature not only expedites the appearance of the perovskite phase but also eliminates solvate intermediates. At 135°C, only the perovskite phase is present. In Figure 5D, SEM images of a blade coating processes conducted at varying temperatures illustrate this phenomenon. At lower temperatures (50°C), the slow evaporation rate leads to the formation of PbI_2 -rich solvate intermediates, resulting in a needle-like

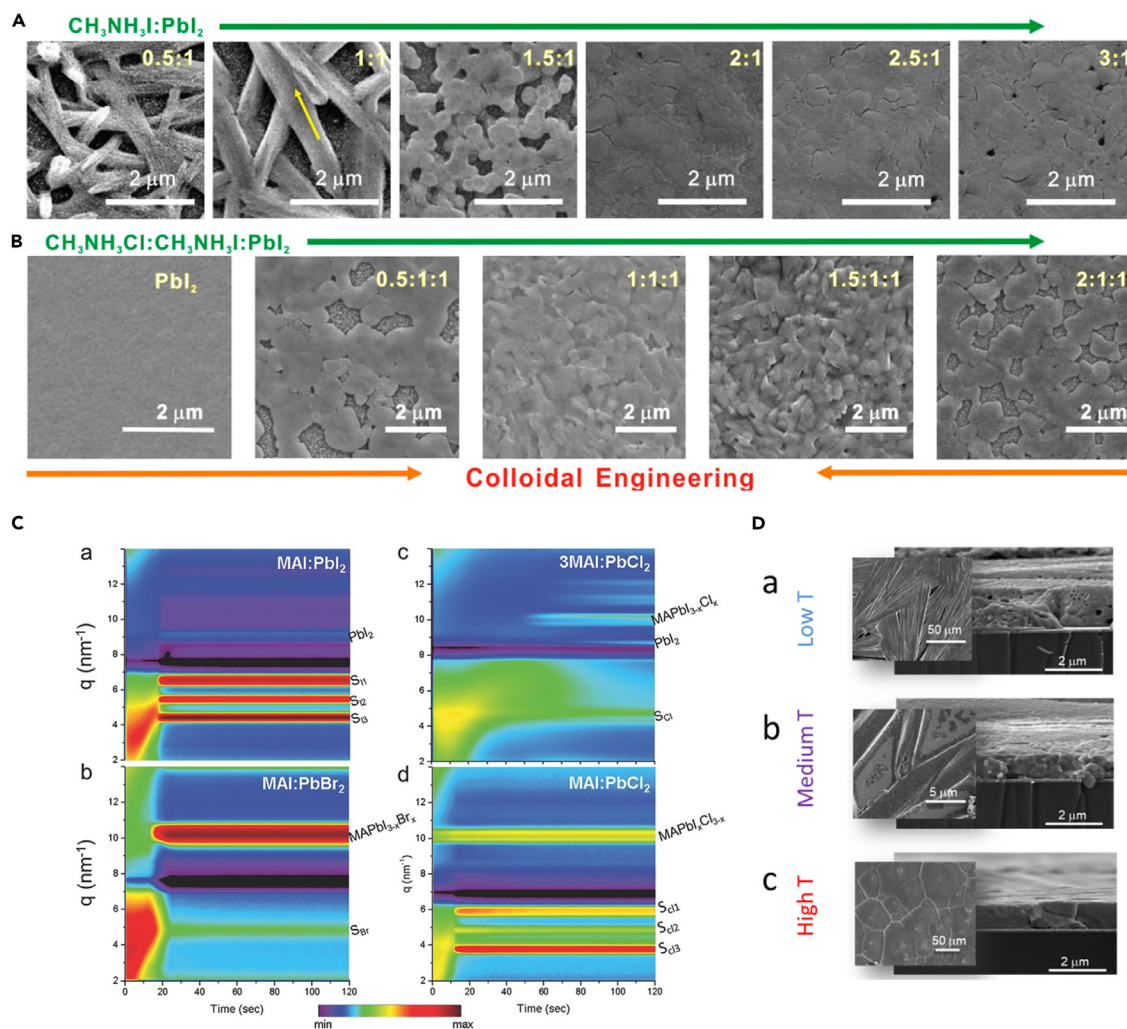


Figure 5. Control on perovskite film morphology

(A) SEM images of PbI_2 thin films with MAI coordinating agent. From left to right, the increase in the MAI: PbI_2 ratio is shown (adapted with permission from Yan et al.⁸¹ Copyright 2015, American Chemical Society).

(B) SEM images of perovskite thin films with MAI coordinating agent. From left to right, the increase in the MAI: PbI_2 ratio is shown (adapted with permission from Yan et al.⁸¹ Copyright 2015, American Chemical Society).

(C) *In situ* GIWAXS measurement during spin coating for different perovskite precursor dispersions: MAI: PbI_2 (a), MAI: PbBr_2 (b), MAI: Cl_2 (c), and 3MAI: Cl_2 (d) (reprinted with permission from Munir et al.⁸²).

(D) SEM images of MAPbI_3 perovskite representing low (a), intermediate (b), and high (c) blade coating temperatures (adapted with permission from Zhong et al.⁸³ Copyright 2018, American Chemical Society).

structure. At intermediate temperatures (between 50°C and 100°C), increased evaporation accelerates the formation of both solvates and perovskite, leading to a competition between these phases. At higher temperatures (above 100°C), only perovskite is formed, yielding films with coarse grains and a high coating degree. High-temperature blade coating processes have demonstrated the ability to achieve remarkably high efficiencies of around 16%–17%.

The control on the film formation is even more difficult in the case of tin-based perovskites, which exhibit considerably faster crystallization rates compared to their lead-based counterparts. One way to act on the crystallization of perovskites is to change the properties of the colloidal precursor suspensions by using the so-called

solution engineering approach. This involves strategic manipulation and optimization of the composition, structure, and processing parameters of perovskite suspensions to enhance the performance and stability of perovskite-based devices.^{11,84,85} This method includes solvent engineering, additive incorporation, and precursor chemistry. By carefully designing and optimizing the precursor composition, the desired properties and characteristics can be achieved in the resulting perovskite films. The classification of a perovskite solution as a chemical solution or a colloidal suspension is still debated, as is the understanding of perovskite formation mechanisms, whether using the classical nucleation and growth model^{86–88} or alternative frameworks such as non-classical nucleation theory.⁴⁶ We believe that seeing perovskite precursor solutions as colloidal suspensions and proposing a formation process based on colloidal interactions would help us better understand perovskite-based systems.

This section is divided into three distinct sub-sections aiming to elucidate the mechanisms through which additives and solvents exert their influence on the colloidal properties of perovskite suspensions and, concurrently, modulate the crystallization dynamics. The categorization of additives and solvents is based on their specific interactions with colloids, resulting in three groups: colloidal repulsion inhibitors (CRIs), colloidal interaction modulators (CIMs), and colloidal stabilizers (CSs). CRIs encompass electrolytic additives that dissolve in the suspension and primarily impact colloids by altering the EDL. CIMs, on the other hand, include electrolytic additives that exhibit chemical and physical interactions with the colloids. Lastly, CSs are molecules capable of forming complexes with colloids and thus stabilizing the colloidal suspension through the mechanism of steric repulsions. The impacts of all additives on colloidal suspension properties and crystallization mechanisms are shown in [Figure 6](#) and summarized in [Table 1](#) for reference and comparison.

CRIs

CRIs primarily consist of salts that dissociate when exposed to a polar solvent, generating positive and negative ions. As delineated in [overview of colloid theory](#), the introduction of a salt into a colloidal suspension prompts the ions to mitigate the charges within the EDL, thereby diminishing electrostatic repulsions. This phenomenon can be demonstrated by referring to the equations in [overview of colloid theory](#). With an increase in the ion concentration within the solution, the EDL thickness diminishes ([Equation 7](#)), consequently diminishing the potential for repulsive forces ([Equation 9](#)). As a result, adding salt causes colloids to aggregate more easily.

Flatken et al. analyzed the impact of A-site cations on the chemistry of MAPbI₃ and FAPbI₃ perovskite suspensions, and the subsequent film formation,⁸⁰ by considering various salts, including potassium, rubidium, and cesium iodide. In [Figure 7A](#), the impact of these cations on the chemistry of the perovskite suspensions is evident through SAXS analysis. Notably, as the concentration of the cations increased, the maximum of the scattering curve became less pronounced and broader. This reduction is attributed to a decrease in the interaction forces among colloids (see [overview of colloid theory](#)). Furthermore, it was observed that the larger the cation, the more pronounced its destabilizing effect on the colloidal suspension, resulting in a greater reduction in the scattering peaks. [Figure 7B](#) displays results from GIWAXS analysis for MAPbI₃ with and without alkali metal additives during annealing in a blade coating process. Noticeable trends were revealed wherein the introduction of additives induces a gradual attenuation of the diffraction peaks occurring at lower theta angles. These peaks are associated with intermediates formed between the perovskite and the solvent. This suggests that the addition of additives destabilizes these

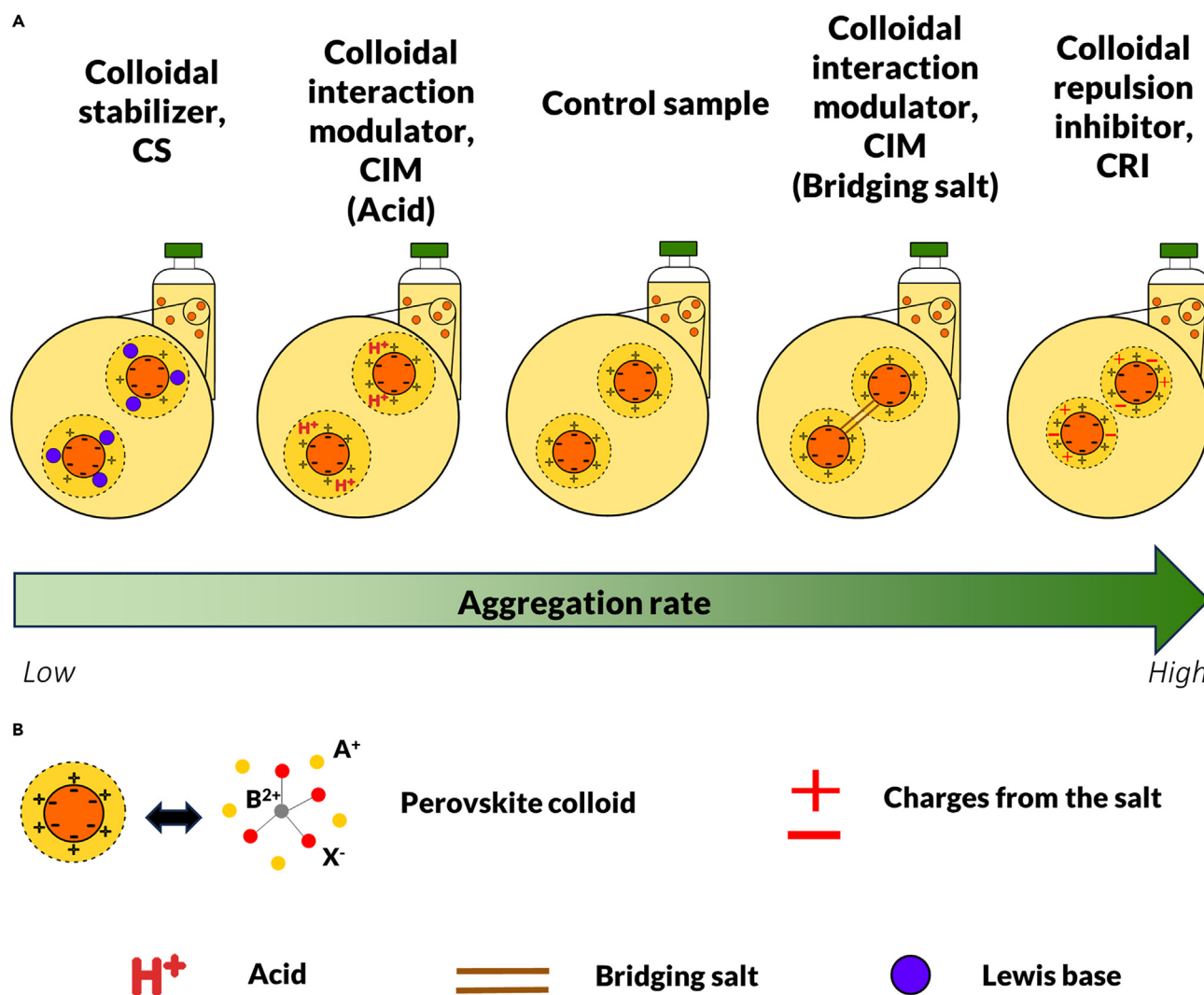


Figure 6. Descriptive diagram showing the effect of different additives

(A) Illustrative figure of perovskite suspensions featuring different additives. The comparative analysis is performed based on the rate of colloid aggregation throughout the perovskite formation process, with the various scenarios compared to the control sample located centrally in the figure. (B) Legend scheme with different elements characterizing perovskite suspensions.

intermediates in favor of perovskite formation. The effect becomes particularly pronounced when transitioning from K^+ and Rb^+ to Cs^+ , with Cs^+ having the largest atomic radius, followed by Rb^+ and K^+ . Pascual et al. likewise observed a comparable phenomenon in their investigation concerning the influence of SnF_2 on $FASnI_3$

Table 1. Summary of the impact of various additives on colloidal suspension properties and crystallization mechanisms

Additive	Molecule type	Effect on colloidal suspension	Effect on the crystallization mechanism
Colloidal repulsion inhibitors (CRIs)	salts	reduced electrostatic repulsions	accelerated aggregation and crystallization
Colloidal interaction modulators (CIMs)	acids bridging salts	increased electrostatic repulsions formation of PNCs	slower aggregation and crystallization PNCs acts as building blocks for facilitated crystallization
Colloidal stabilizers (CSs)	Lewis bases	increased colloid repulsions due to the volumetric hindrance	decrease in the aggregation process and slower crystallization

PNCs, pre-nucleation clusters.

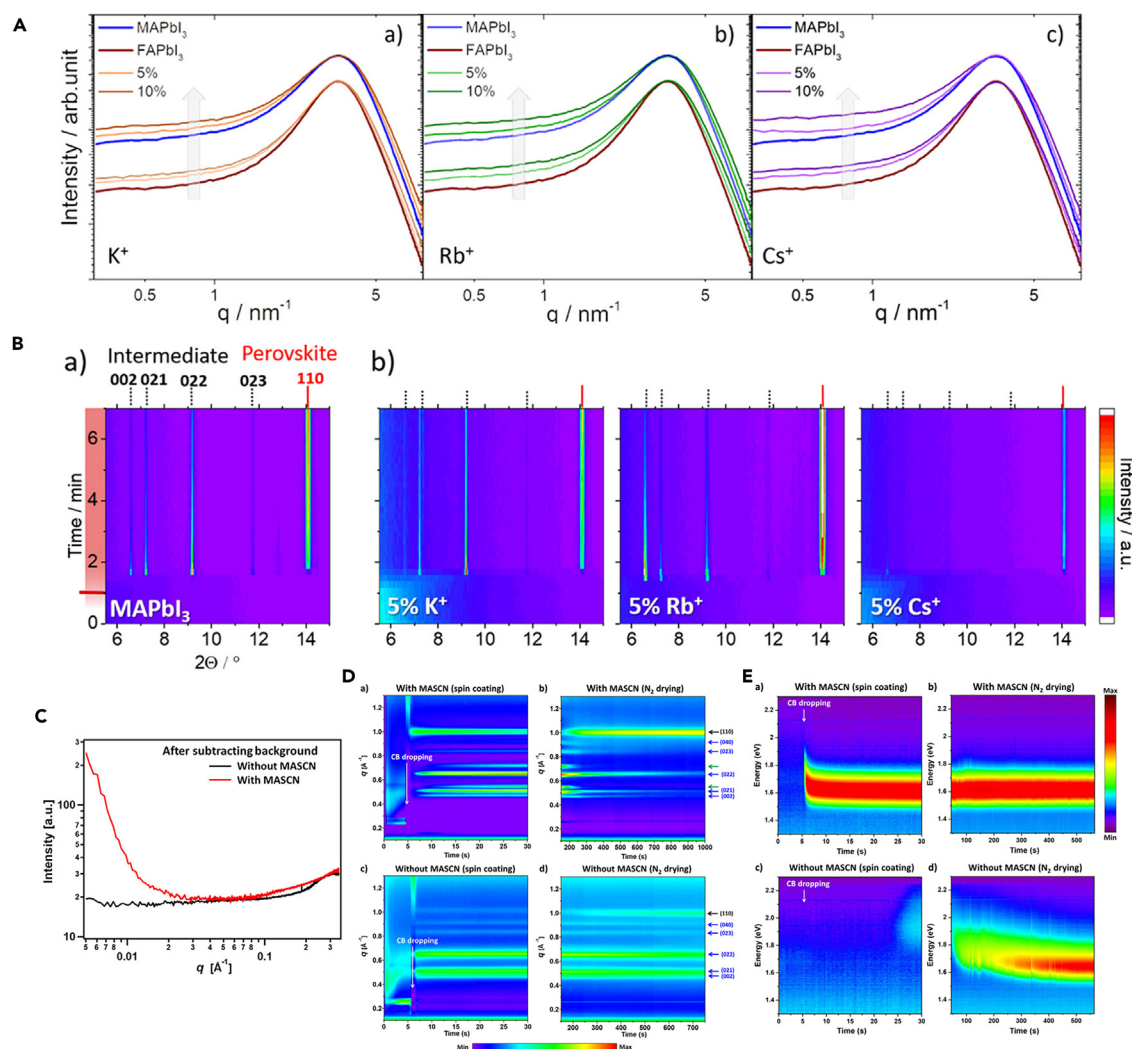


Figure 7. Evaluation of the CRIs on lead perovskite formation

(A) SAXS data on MAPbI₃ and FAPbI₃ precursor dispersions with different concentration of K⁺ (a), Rb⁺ (b), and Cs⁺ (c) (adapted with permission from Flatken et al.⁸⁰ Copyright 2022, American Chemical Society).

(B) *In situ* GIWAXS data during annealing. (a) represents the pristine sample, and (b) represents the dispersions with the addition of K⁺, Rb⁺, and Cs⁺, respectively (adapted with permission from Flatken et al.⁸⁰ Copyright 2022, American Chemical Society).

(C) SAXS measurement for MAPbI₃ in DMF/DMSO without the addition of MASCN (black curve) and with the addition of MASCN (red curve) after background removal (adapted with permission from Abdelsamie et al.⁸⁹ Copyright 2021, American Chemical Society).

(D) *In situ* GIWAXS during perovskite film formation (spin coating and N₂ drying) with the addition of MASCN (a and b) and without the addition of MASCN (c and d) (reprinted with permission from Abdelsamie et al.⁸⁹ Copyright 2021, American Chemical Society).

(E) *In situ* PL during perovskite film formation (spin coating and N₂ drying) with the addition of MASCN (a and b) and without the addition of MASCN (c and d) (reprinted with permission from Abdelsamie et al.⁸⁹ Copyright 2021, American Chemical Society).

tin-based perovskites.⁹⁰ By analyzing SAXS data, they noted that the introduction of SnF₂ had the capacity to reduce the intensity of the structure factor peak. Furthermore, it led to a notable shift of this peak toward higher values of the scattering vector q , indicating a reduction in the interparticle spacing.

Abdelsamie et al. studied the impact of methylammonium thiocyanate (MASCN) salt on the MAPbI₃ perovskite.⁸⁹ Their study comprised two primary aspects: the characterization of the perovskite precursor suspension and the investigation of crystallization dynamics. To gain insights into the properties of the suspension, SAXS

experiments were performed, and their outcomes are reported in [Figure 7C](#). It was observed that the addition of 50% MASCN led to a substantial increase in the scattering curve at low q compared to the sample without the additive. The resulting negative slope of the curve is linked to an enlargement in the size of aggregates in the suspension. This finding corroborates the results obtained by Han et al. in 2017,⁹¹ who performed DLS experiments on MAPbI₃ dispersions containing MASCN. For the assessment of crystallization dynamics, *in situ* GIWAXS ([Figure 7D](#)) and photoluminescence (PL; [Figure 7E](#)) tests were carried out during the spin coating process, followed by drying under N₂ atmosphere. GIWAXS analysis revealed that, in the presence of the additive, the main perovskite's diffraction peaks emerged immediately upon the application of the anti-solvent, while the diffraction peaks associated with perovskite-solvent intermediates diminished. In contrast, in the absence of the additive, perovskite formation was significantly delayed and occurred later during the drying phase. Consequently, MASCN demonstrated a propensity to accelerate perovskite formation while destabilizing perovskite-solvent intermediates. This observation was consistent with *in situ* PL results, wherein the sample containing the additive exhibited PL immediately after anti-solvent application, while the pristine sample's PL manifested toward the end of the spin coating process.

The influence of the MASCN additive was also examined by Zhang et al. in 2022 in the context of two-dimensional (2D) perovskites.⁹² The authors employed this additive as a mean to regulate the crystallization process within the Ruddlesden-Popper (RP) 2D perovskites (PA)₂(MA)₄Pb₅I₁₆.⁹³ Much like the earlier case, there is a discernible trend of increasing colloid size (DLS measurement) corresponding to higher additive concentrations, which promotes the generation of a film characterized by larger grains and increased crystallinity.⁹⁴

Li et al. analyzed the influence of MACl salt on MAPbI₃ perovskite, aiming at enhancing control over the material's morphology.⁹⁵ In [Figure 8A](#), the authors illustrated their conceptual framework, elucidating the mechanistic foundations governing perovskite film formation from the suspension. The formation of a multilayer gel wet film occurs when colloidal cluster sizes are smaller than the wet film's thickness. Conversely, when cluster sizes are close to the wet film's thickness, systematic arrangement on the substrate leads to the creation of a monolayer wet gel film with an improved film quality and material properties after the residual solvent removal. The researchers utilized MACl salt as an aggregating additive for the colloidal suspension. Through DLS, they observed a proportional increase in colloid size with higher salt concentrations, as depicted in [Figure 8B](#), resulting in an increase in grain size, as illustrated by the SEM images reported in [Figure 8C](#). Specifically, the average grain size expands from 230 nm to 3 μ m with the addition of MACl. In a recent study by Shin et al., the same additive was employed in the context of a FAPbI₃ perovskite.⁸⁷ Consistent with the observations made by Li et al.,⁹⁵ the MACl salt prompts the generation of larger colloids within the suspension. Furthermore, the authors elucidated the impact of this additive on crystal orientation. As depicted in [Figure 8D](#), grazing-incidence X-ray diffraction (GIXRD) analysis was carried out on samples containing different concentrations of MACl, specifically 10%, 20%, 30%, and 40%, revealing that the sample containing 40% MACl exhibited the highest intensity in the out-of-plane direction in comparison to the others. In contrast, the samples with lower MACl concentrations displayed Scherrer's ring patterns indicative of a random orientation distribution.⁹⁴

Spacer cations, apart from their involvement in the genesis of 2D perovskites, manifest alterations in the colloidal attributes of perovskite dispersions when

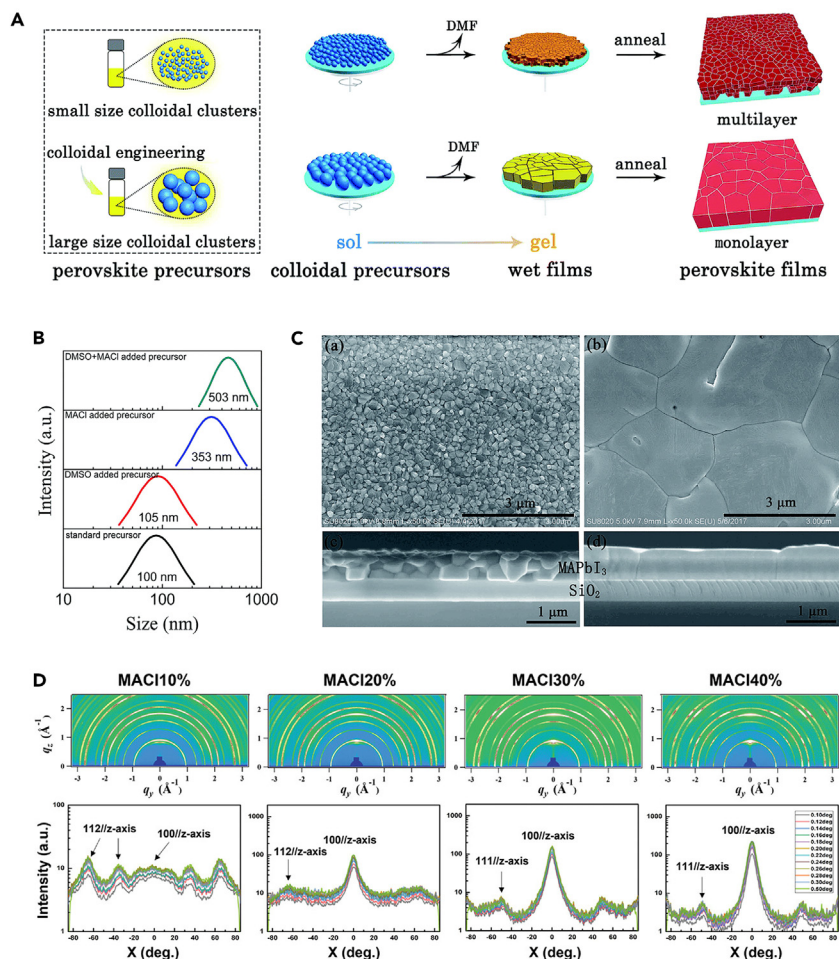


Figure 8. Analysis of MACI's effect on lead perovskites

(A) Schematic representation of the perovskite film formation process originating from the precursor suspension. The MAPi precursor salts are dispersed in DMF and DMSO (reproduced from Li et al.⁹⁵ with permission from the Royal Society of Chemistry).

(B) DLS analysis of different perovskite precursor suspensions (reproduced from Li et al.⁹⁵ with permission from the Royal Society of Chemistry).

(C) Surface SEM and cross-sectional SEM images of the annealed perovskite film without MACI (a) and with MACI (b) (reproduced from Li et al.⁹⁵ with permission from the Royal Society of Chemistry).

(D) Synchrotron-based GIXRD patterns of FAPbI₃ film with different content of MACI (top row). Azimuthally integrated 1D plots along 100 plane (bottom row) (adapted with permission from Shin et al.⁹⁶).

introduced as salts. Pan et al. investigated the influence of different spacer cations on the properties of RP tin-based perovskite films, specifically focusing on the 2-thiophenemethylammonium cation (TH⁺) and the benzylamine cation (BE⁺).⁹⁷ They proposed an effective strategy to attenuate the kinetics of perovskite crystallization by introducing competitive intramolecular and intermolecular interactions. Their investigation revealed that the TH⁺ cation, due to the presence of sulfur in its heterocycle, exhibited greater stability compared to the BE⁺ cation. Hence, in contrast to colloids based on BE⁺ cations, those relying on TH⁺ cations exhibit enhanced electrostatic repulsion, effectively impeding particle aggregation. Figure 9A (a) illustrates the results of zeta potential measurements, showing that TH₂FA₃Sn₄I₁₃ exhibits a higher zeta potential. Furthermore, the authors performed

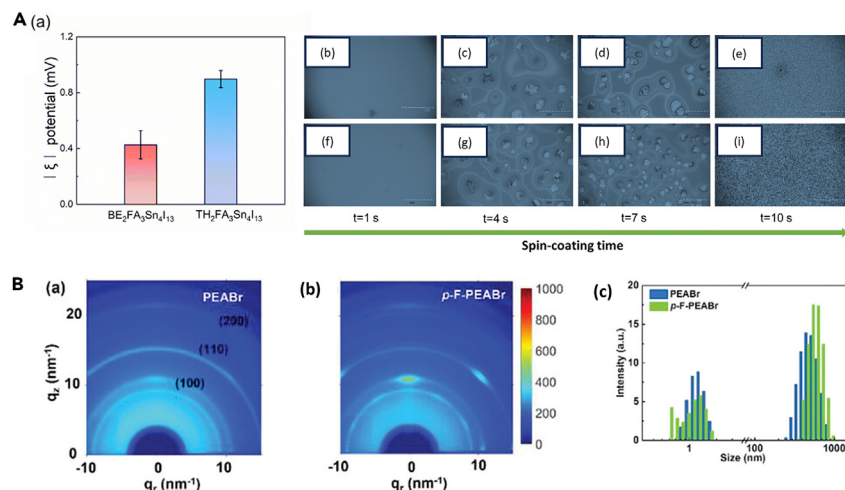


Figure 9. Study of CRI spacer cations on 2D perovskites

(A) Zeta potential measurement for $\text{BE}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ and $\text{TH}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ perovskite materials (a) and *in situ* optical microscopy during spin coating for $\text{BE}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ (b–e) and $\text{TH}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ (f–i) (reproduced from Pan et al.⁹⁷ with permission from the Royal Society of Chemistry). (B) GIWAXS images for PEABr (a) and p-F-PEABr (b) quasi-2D perovskites and DLS analysis of PEABr and p-F-PEABr dispersions (c) (adapted with permission from He et al.⁹⁸).

optical microscopy observations during the spin coating process, also shown in Figure 9A (b–i). They noted that, in the case of $\text{BE}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ (Figure 9A, panels b–e), the formation of aggregates occurred in a relatively unstable manner, resulting in an irregular film morphology characterized by the presence of voids. Conversely, with $\text{TH}_2\text{FA}_3\text{Sn}_4\text{I}_{13}$ (Figure 9A, panels f–i), the aggregates exhibited a more even dispersion, facilitating the creation of a film with greater homogeneity.

He et al. investigated the impact of spacer cations on quasi-2D perovskites, specifically focusing on CsPbI_3 materials,⁹⁸ which are appreciated for their remarkable stability. However, their layered structure poses challenges related to charge transfer, consequently limiting their PCE. In addressing this issue, the researchers conceived the notion of employing salts with the capacity to form larger colloidal particles as spacer cations. These larger colloids can serve as nuclei for crystallization, promoting the process. In their study, two different salts were used, i.e., PEABr and p-F-PEABr. The only difference between the two salts lies in the presence of fluoride in the *para* position of p-F-PEABr. This structural disparity results in higher instability compared to the simple PEABr, which is evident both in the dispersion properties and in the crystalline structure. Consistent with this theoretical framework, Figure 9B shows higher Bragg spot in the GIWAXS images for p-F-PEABr compared to those for PEABr (as depicted in Figure 9B, panels a and b). Furthermore, the analysis of DLS data reveals that the colloids exhibit a larger size in the presence of the fluorinated salt, as depicted in Figure 9B (c).

CIMs

The CIM category covers all electrolytic additives that are capable of dissolving in solution and are also capable of chemically interacting with perovskite colloids, as, for instance, the acids and the bridging salts. Regarding the acids, the zeta potential tends to be positive under acidic pH conditions and negative in alkaline environments. This trend signifies that colloid stability follows a parabolic pattern, with a minimum at the point of zero zeta potential, specifically the isoelectric point.

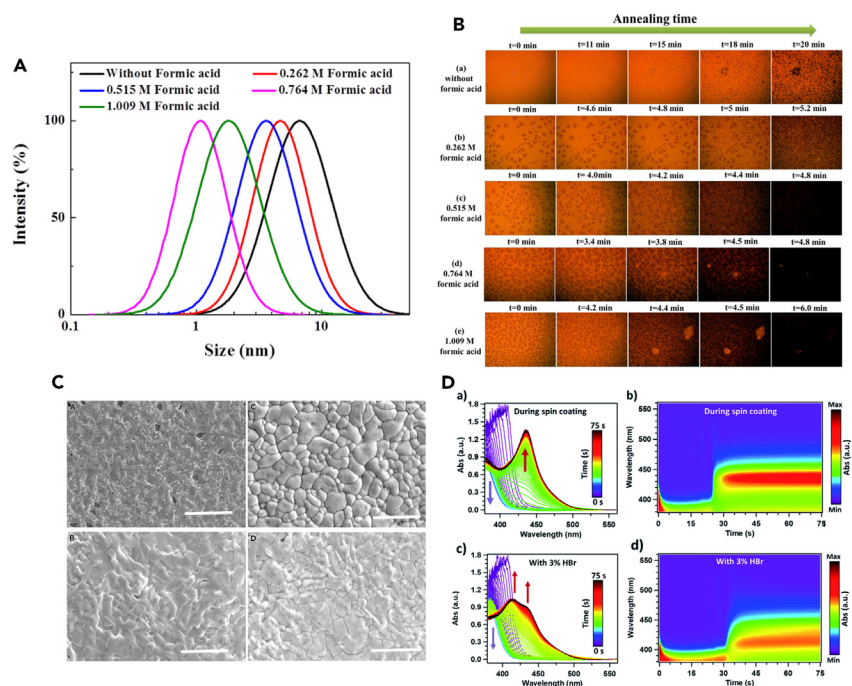


Figure 10. Analysis of CIM acids on perovskite materials

(A) DLS analysis of FAPbI₃ dispersions with different amounts of formic acid (adapted from Meng et al.⁹⁹ Copyright 2020, with permission from Elsevier).

(B) *In situ* microscopy during crystallization of FAPbI₃ without (a) and with (b–e) formic acid (reprinted from Meng et al.⁹⁹ Copyright 2020, with permission from Elsevier).

(C) Surface SEM images of MAPbI_{3-x}Cl_x films with different formic acid concentrations: 0 (a), 3 (b), 5 (c), and 10 vol % (d) (adapted from Noel et al.¹⁰⁰ Copyright 2020, with permission from Elsevier).

(D) *In situ* UV-vis analysis during spin coating of Cs₂AgBiBr₆ precursor suspensions without (a and b) and with (c and d) HBr. The graphs on the left are the absorbance spectra, while the graphs on the right are the contour maps (reproduced from Abdelsamie et al.¹⁰¹ with permission from the Royal Society of Chemistry).

This behavior stands in stark contrast to that of simple salts, where the zeta potential monotonically decreases to zero with increasing electrolyte concentration.⁴⁹ Consequently, it can be inferred that an elevation in acid concentration correlates with an increase in the stability of the colloidal suspension. Acids are commonly employed to diminish the size of aggregates within the perovskite suspension by fostering repulsive interactions among colloids.

Meng et al. investigated the impact of formic acid addition to a FAPbI₃ perovskite suspension.⁹⁹ As shown in Figure 10A, they observed that the incorporation of formic acid leads to a reduction in the colloidal particle size and an increase in their zeta potential. The authors assumed that a reduction in the dimensions of the colloidal particles might result in an increased colloidal density, consequently yielding a more densely packed film. By employing *in situ* microscopy to examine the crystallization dynamics during annealing (Figure 10B), they found that the presence of formic acid facilitated the formation of smaller yet denser nuclei, resulting in an improved film coverage. In contrast, the absence of the acid yielded larger and more dispersed starting nuclei. A similar result was obtained by Noel et al., who studied the impact of formic acid on mixed-halide perovskites MAPbI_{3-x}Cl_x.¹⁰⁰ Their study revealed that precise manipulation of pH levels can induce significant changes in the colloidal distribution within the suspension. Through DLS, they observed that

in the absence of the acid, colloids of different sizes were present, whereas with the addition of formic acid into the precursor suspension, the presence of larger colloids was entirely eliminated, leading to a more uniform distribution of colloids in the suspension. Figure 10C illustrates that even a minor addition of formic acid, such as 3%, can lead to the formation of a higher-quality film characterized by a more regular morphology.

Abdelsamie et al. analyzed the impact of hydrobromic acid (HBr) on the fabrication process of lead-free perovskite $\text{Cs}_2\text{AgBiBr}_6$.¹⁰¹ As shown in Figure 10D, the analysis of the material without the additive via ultraviolet-visible (UV-vis) spectroscopy during the spin coating unveiled the presence of an absorbance peak, typically between 420 and 440 nm (Figure 10D, panel a). The introduction of HBr as an additive led to the emergence of a new peak at approximately 410 nm, a phenomenon the authors attributed to the formation of an HBr-perovskite complex. Crucially, it should be noted that the formation of the complex appeared to retard the evolution of the peak at 439 nm. The observed correlation between the additive and the dynamics of perovskite formation was further corroborated by DLS measurements, where the authors observed that HBr induced a shift in the size distribution of colloids toward smaller values, substantiating the hypothesis that the additive increased repulsive forces between colloids, thus stabilizing the perovskite precursor suspension. This assertion is confirmed by their GIWAXS data, indicating that diffraction peaks associated with the additive emerge later compared to the case without additives. McMeekin et al. studied the combined effect of HBr and hydroiodic acid on perovskites with the composition $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$.⁷⁸ In their study, they investigated the impact of additives on the aging process of the perovskite precursor suspension. DLS was employed to monitor temporal changes in colloidal particle sizes. The findings revealed that with prolonged aging, colloidal particles exhibited a reduction in size. This reduction in particle size was corroborated by a decrease in the suspension's turbidity, as evidenced by the Tyndall effect. Additionally, the study incorporated *in situ* microscopy during the annealing of the perovskite film. The observations indicated that suspensions containing larger colloids (without additives) underwent rapid growth during annealing. In contrast, the introduction of acids, coupled with extended aging time, prolonged the growth process, resulting in the formation of polycrystalline films with larger grain structures.

While many studies have predominantly explored the impact of acids on 3D perovskites, it is noteworthy that certain acids also serve as regulators of crystallization in the context of 2D perovskite materials. For instance, a study by He et al. delves into the effects of 2-aminophenol-4-sulfonic acid (APSA) on meta-(aminomethyl) piperidinium quasi-2D lead perovskites.¹⁰² In the 2D perovskites field, a key objective is to mitigate the wide distribution of the numbers of unit cells per layer "*n*," a parameter that significantly influences device performance. One approach to achieve this is by eliminating complex aggregates within the suspension. Through the addition of APSA, the authors successfully accomplished a reduction in the size of colloidal aggregates within the dispersion, as is evident from the DLS results. This reduction in colloidal size corresponds to a decrease in the rate of crystallization, as observed from the temporal evolution of absorbance during the spin coating process. Furthermore, analysis via GIWAXS revealed that at a photon beam incidence angle of 0.1° , untreated perovskites exhibited the presence of *n* values of 2 and 3. In contrast, perovskites treated with AMPA displayed solely an *n* value of 3.

Bridging salts represent another subtype of additives within the category of CIMS. They possess the capability to alter the size distribution of the perovskite suspension

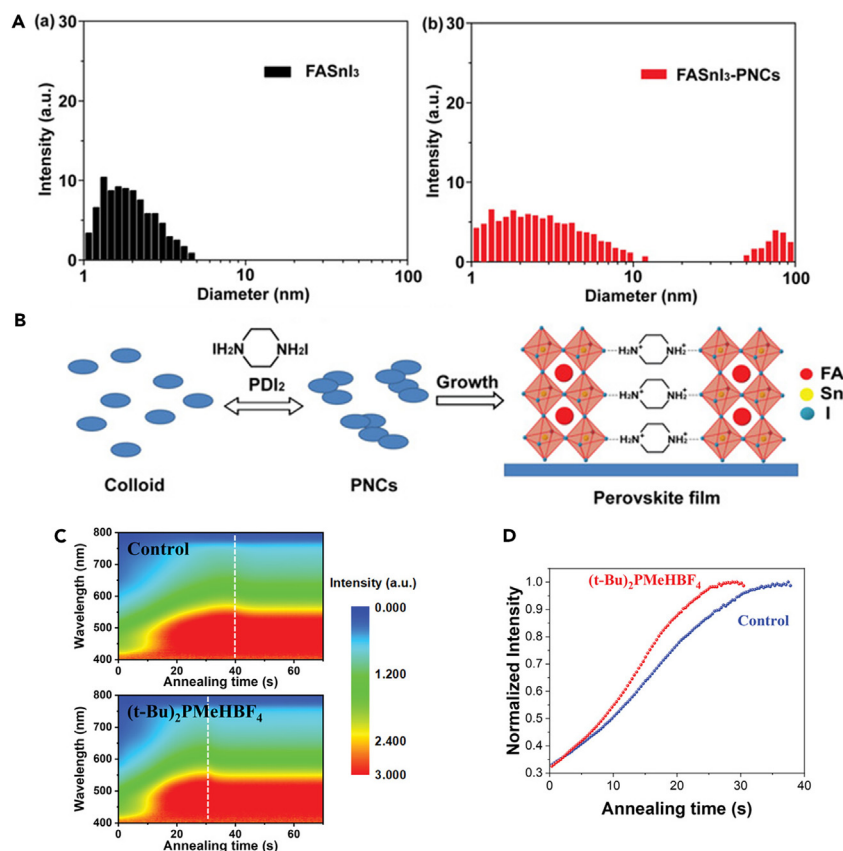


Figure 11. Pre-nucleation cluster formation induced by CIMs

(A) DLS analysis of the FASnI₃ dispersion without PDI₂ (a) and with (b) PDI₂ (adapted with permission from Meng et al.¹⁰³).
 (B) Illustration of the perovskite formation process with PNCs (adapted with permission from Meng et al.¹⁰³).
 (C) *In situ* UV-vis contour maps for the annealing process for the control perovskite film and for the film with 0.3 mol % (t-Bu)₂PMeHBF₄ (adapted with permission from Zhang et al.¹⁰⁴).
 (D) Absorption intensities calculated at the wavelength of 550 nm during the annealing (adapted with permission from Zhang et al.¹⁰⁴).

toward larger values by forming bridging structures. These structures, known as pre-nucleation clusters (PNCs), serve as building blocks or seeds to enhance the crystallization process. Meng et al. explored the influence of the piperazine dihydriodide (PDI₂) on tin-based perovskite colloidal suspensions.¹⁰³ They observed a notable transformation of the colloidal properties due to the presence of piperazine. The authors identified the formation of PNCs as a significant outcome of this interaction. The PNCs were deemed to act as intermediate species during the perovskite crystallization process, ultimately leading to the development of high-quality films characterized by low defect densities. To understand the impact of piperazine, DLS measurements were performed on both the pure suspension and the suspension with the additive, as shown in Figure 11A. The results demonstrated that the pristine sample displayed a colloidal distribution centered around 2.9 nm, whereas the introduction of piperazine induced the emergence of a bimodal distribution with peaks centered at 5.2 and 81 nm, with the latter corresponding to the PNCs. The authors speculated that these larger clusters might serve as building blocks in the perovskite formation process, facilitating the crystallization, as depicted in Figure 11B. The enhanced film quality was further corroborated through XRD analysis, which

revealed more intense peaks, and SEM images, indicating a more uniform distribution of grains.

Analogously, Zhang et al.¹⁰⁴ focused on the impact of the ionic salt (t-Bu)₂PMeHBF₄ on a triple-cation mixed-halide perovskite composite, denoted as (FAPbI₃)_{0.93}(MAPbBr₃)_{0.04}(CsPbI₃)_{0.03}. Typically, perovskite thin-film formation processes with slow crystallization dynamics result in non-uniform structures with limited coverage. In line with established practices in the semiconductor industry, the authors introduced seed materials to obtain high-quality products. In this regard, the employed salt served as seed generators to accelerate the crystallization process. Through DLS analysis, it is notably observed that the addition of the salt induces an enlargement in the size of aggregates within the suspension. To illustrate the intimate connection between colloidal suspension properties and film crystallization, the researchers performed *in situ* UV-vis experiments during the annealing process at 70°C for samples both with and without the additive. The sample containing the additive attains a steady-state absorbance at almost 20 s, whereas the sample without the additive does not, as demonstrated in [Figures 11C and 11D](#). This result is consistent with the observations made by *in situ* microscopy. Specifically, after 20 s of annealing, the sample containing the additive exhibited a significant number of crystals, in contrast to the pure sample, where only a limited number of crystals were observed.

Su et al. investigated the influence of acetic acid (HAc) on tin-based perovskites¹⁰⁵ and made an interesting observation that sets it apart from other acids,⁹⁸ but it is consistent with the last additives we mentioned. In contrast to the effects provided by other acids discussed above, the introduction of HAc induces the formation of aggregates that expedite the crystallization process. DLS measurements clearly demonstrate a significant shift in size distribution toward larger values. This shift is evident as the colloidal particle size increases from 376 nm without HAc to 704 nm with 15% HAc. Surface morphology analysis through SEM provides unequivocal evidence of the distinctive impact of this additive. The sample without the additive shows pinholes, whereas the sample with HAc not only removes such imperfections but also displays coarser grain structures. To comprehensively elucidate differences in crystal structure, XRD analyses consistently reveal lower full width at half maximum (FWHM) values and higher intensities in diffraction peaks in the presence of the additive compared to the control sample, indicative of improved crystalline characteristics. Similar findings were also obtained in a study performed by Wang et al. in which they examined the impact of formamidine acetate (FAAc) compared to HAc and methylammonium acetate (MAAc).¹⁰⁶

CSs

This section examines the class of compounds that, through Lewis base-acid interaction, can form complexes with perovskite colloids. As elucidated in [overview of colloid theory](#), when a molecule comparable in size or larger than the colloid binds to it, it creates a shielding effect. In practical terms, colloids encountering one another must initially surpass the repulsive force arising from these molecules, a phenomenon known as steric repulsion. Consequently, the presence of such molecules substantially diminishes colloid aggregation and influences the perovskite crystallization mechanism.

DMSO is a frequently employed solvent in the fabrication of perovskite-based solar cells due to its high donor number, which enables it to exert control over their intricate crystallization processes. In the work from Li et al.,¹⁰⁷ the influence of DMSO was investigated within a complex precursor system involving various salts, including

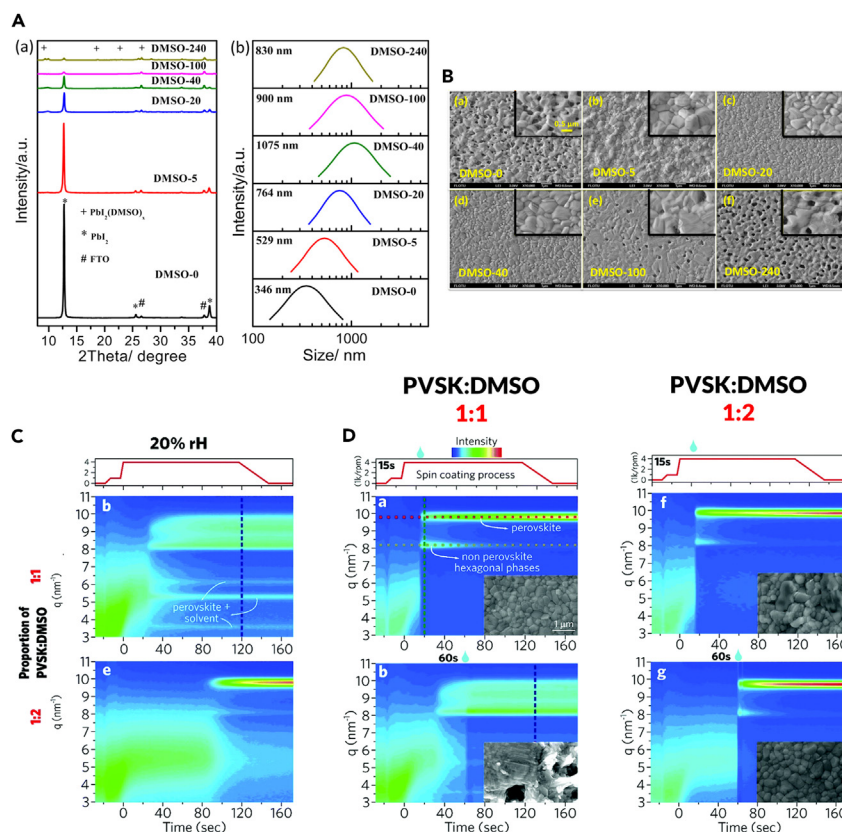


Figure 12. Role of DMSO on the crystallization process of lead perovskites

(A) XRD analysis of PbI_2 -DMSO films with increasing volumes of DMSO (a) and DLS analysis of PbI_2 -DMSO complexes (b) (adapted with permission from Li et al.¹⁰⁷ Copyright 2015, American Chemical Society).

(B) SEM images of perovskite thin films with increasing volumes of DMSO (reprinted with permission from Li et al.¹⁰⁷ Copyright 2015, American Chemical Society).

(C) *In situ* GIWAXS during spin coating of CsFA perovskite precursor suspensions with a 1:1 perovskite (PVSK):DMSO ratio (b) and 1:2 PVSK:DMSO ratio (e) at 20% relative humidity (reproduced from Szostak et al.¹⁰⁸ with permission from the Royal Society of Chemistry).

(D) *In situ* GIWAXS during spin coating with anti-solvent dripping of CsFA suspension with different ratios of PVSK:DMSO. (a) and (f) are the contour maps with the dripping at 15 s, and (b) and (g) are the contour maps with the dripping at 60 s (reproduced from Szostak et al.¹⁰⁸ with permission from the Royal Society of Chemistry).

methylammonium iodide (MAI), methylammonium bromide (MABr), formamidinium iodide (FAI), and formamidinium bromide (FABr), along with DMF as a solvent. The perovskite fabrication process adopted in this study employed a dual-step sequential deposition method. Initially, a dispersion containing lead iodide PbI_2 in DMSO and DMF was prepared and subsequently spin coated, followed by the addition of the organic cation salt. The investigation started with an analysis of the PbI_2 -DMSO suspension, where DLS revealed that an increased percentage of DMSO led to larger colloidal particles, as shown in Figure 12A (b). The increase in the colloid size may be attributed to the reduction in the colloid diffusion coefficient. Simultaneously, the crystalline characteristics of the film after spin coating were assessed, as illustrated in Figure 12A (a). Notably, an increased DMSO content is correlated with a reduction in the intensity of the peak corresponding to the PbI_2 phase. This observation is indicative of intricate complexing effects within the system. These findings were closely linked to the final morphology of the film obtained through

the dual-step deposition method (Figure 12B). Specifically, larger colloids were associated with coarser and more crystalline grains compared to those in the additive-free scenario.¹⁰⁹

Szostak et al. investigated the impact of DMSO on the crystallization dynamics of $\text{Cs}_{0.17}\text{FA}_{0.83}\text{Pb}(\text{Br}_{0.17}\text{I}_{0.83})_3$ perovskite.¹⁰⁸ Their research aimed to understand the influence of the DMSO:DMF solvent ratio on the perovskite crystallization process. The study employed *in situ* GIWAXS to monitor the spin coating process. Initially, without the application of an anti-solvent, the 2D contour maps exhibited a broad scattering signal attributed to the presence of a colloidal gel, as depicted in Figure 12C. Subsequently, this gel phase transitioned to reveal the characteristic perovskite and intermediate phase peaks. Notably, the duration of the colloidal gel phase was found to vary based on the DMSO:DMF ratio. In particular, an increase in the DMSO:DMF ratio was associated with an extended lifetime of the colloidal gel phase. In Figure 12C (b), the transition from a gel-like state to a solid film occurs at approximately 30 s when the precursor suspension has a 1:1 perovskite:DMSO ratio, whereas in Figure 12C (e), this transition takes place at around 80 s with a ratio of 1:2. An essential aspect for achieving high-quality films was the use of the anti-solvent and its application time. For instance, when the perovskite:DMSO ratio was 1:1, applying the anti-solvent at 15 s led to the formation of a well-structured film devoid of hexagonal phases at the end of the spin coating, as depicted in Figure 12D (a). In contrast, Figure 12D (b) shows that applying the anti-solvent at 60 s, after the colloidal gel phase had dissipated, resulted in an irregular film characterized by large pinholes and a notable presence of hexagonal-phase lead iodide PbI_2 . However, when the DMSO:DMF ratio was 2:1, the application of the anti-solvent at either 15 or 60 s (Figure 12D, panels f and g, respectively) yielded favorable outcomes, as the colloidal gel phase was still present during the anti-solvent application in both scenarios. This underscores the critical importance of ensuring the presence of the complexed phase, namely, the colloidal gel phase, at the time of anti-solvent application to achieve films with uniform and homogeneous morphologies. The retarding effect of DMSO on colloidal gel was also seen by Shargaieva et al. 1 year later on the MAPbI_3 perovskite.¹¹⁰

The formation of MAPbI_3 -DMSO complexes can be facilitated by the introduction of molecules containing Lewis functional groups during the anti-solvent application process. In a study conducted by Niu et al., molecules such as 3,9-bis(2-methylene-(3-(1,1-dicyanomethylene)-indanone))-5,5,11,11-tetrakis(4-hexylphenyl)-dithieno [2,3-d':2',3'-d'']-s-indaceno[1,2-b:5,6-b']dithiophene (ITIC) and 5z,5'e)-5,5'-(((4,8-bis (5-(2-ethylhexyl)thiophen-2-yl)benzo[1,2-b:4,5-b']dithiophene-2,6-diyl)bis(3,3''-diocetyl-[2,2':5',2''-terthiophene]-5'',5-diyl))bis(methaneylylidene))bis(3-ethyl-2-thioxothiazolidin-4-one (DR3TBDTT) were introduced into chlorobenzene, serving as the anti-solvent.¹¹¹ Comparative analysis with pure anti-solvent revealed that these molecules not only enhance the formation of MAPbI_3 -DMSO complexes but also orient their growth, as evidenced by GIWAXS analysis.

Lee et al. studied the influence of other CSs on lead-based perovskites.¹¹² Specifically, they investigated the impact of four distinct 2-pyrrolidone additives, N-methyl-2-pyrrolidone, N-ethyl-2-pyrrolidone, N-cyclohexyl-2-pyrrolidone (CHP), and N-octyl-2-pyrrolidone, while the main solvent was DMF. Their molecular structures are shown in Figure 13A. These solvents possess different physical properties, but they share the common characteristic of being strong electron donors exhibiting Lewis base behavior. Upon thermal annealing and SEM examination, the authors found out that, in the absence of additives, the film's morphology was characterized

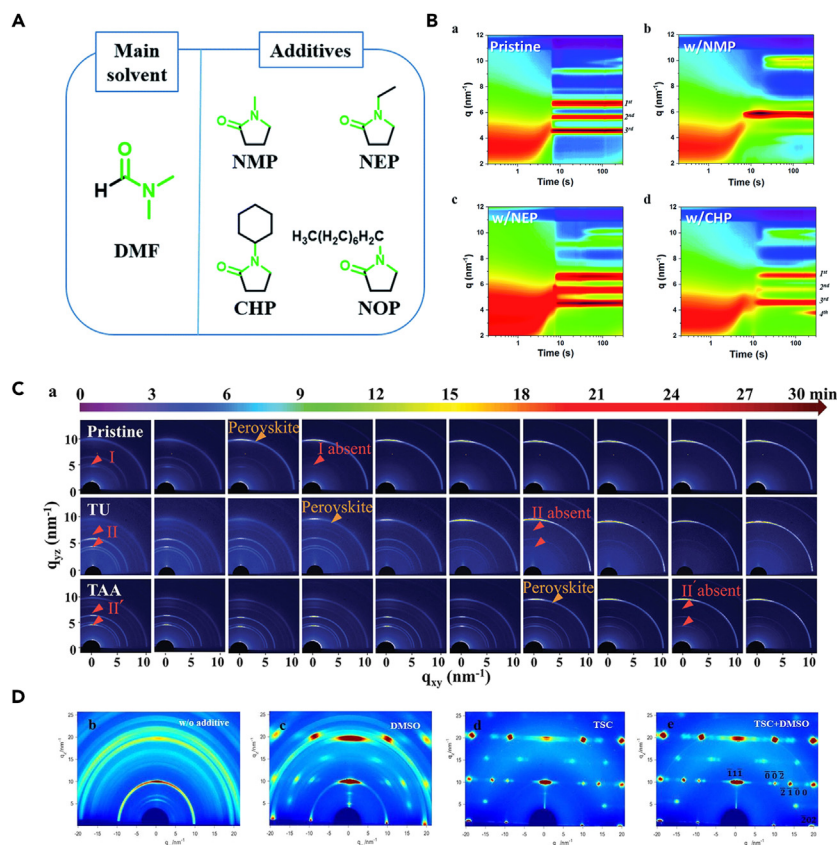


Figure 13. Evaluation of the effect of different CSs on 3D and 2D lead perovskites

(A) Molecular structures of the main solvent and additives used by Lee et al.¹¹² (reproduced from Lee et al.¹¹² with permission from the Royal Society of Chemistry).
 (B) *In situ* GIWAXS contour maps during spin coating of different perovskite dispersions (reproduced from Lee et al.¹¹² with permission from the Royal Society of Chemistry).
 (C) *In situ* 2D GIWAXS maps during the annealing process of the pristine, TU-based, and TAA-based perovskite films (adapted with permission from Xu et al.¹¹³).
 (D) GIWAXS patterns of the crystallized films without (b) and with (c–e) the additive (adapted from Zheng et al.¹¹⁴ Copyright 2020, with permission from Elsevier).

by needle-like structures, mainly composed of PbI_2 , with large pinholes. However, the addition of the aforementioned solvents as additives led to a notable enhancement in film morphology. It is worth mentioning that the sample with CHP showed a complete absence of pinholes. Figure 13B depicts the GIWAXS results during spin coating. It became evident that the presence of the additives delayed the appearance of diffraction peaks compared to the pristine sample. Three distinct diffraction peaks at 6.8, 5.7, and 4.6 nm⁻¹ were observed in the pristine sample, with the central peak at 5.7 nm⁻¹ attributed to the PbI_2 -DMF phase. Interestingly, the addition of CHP to the suspension resulted in the suppression of this peak, indicating that CHP possessed a higher complexation ability, compared to DMF, in forming adducts with PbI_2 .

CSs have found use in the field of 2D perovskites. For instance, thiourea (TU) and thioacetamide (TAA) were employed as complexing agents for lead-based perovskites in a study performed by Xu et al. in 2021.¹¹³ Both molecules, characterized by an electron donor group = S and two -NH₂ groups capable of forming hydrogen bonds, exhibit interactions with the perovskite leading to complex formation. The

investigation focused on their impact on the annealing process of MAPbI₃-type perovskite utilizing synchrotron-based *in situ* GIWAXS. At the end of the annealing process (Figure 13C, first row), scattering rings at $q = 4.4$, 4.9 , and 6.3 nm^{-1} , attributed to a phase comprising DMSO and PbI₂, were observed and subsequently vanished after approximately 6 min, giving way to the appearance of the characteristic perovskite ring at 10 nm^{-1} . When TU or TAA additives were introduced (Figure 13C, second and third row), new phases II and II' emerged at $q = 8.2$ and 9.3 nm^{-1} , respectively, in addition to the previously identified ones associated with perovskite-TU/TAA-solvent complexes. Notably, the durations of these phases were distinct compared to phase 1 in the control sample, persisting for 9 min with TU and 18 min with TAA. Following the disappearance of such additional phases, perovskite formation ensued. This elucidates how TU and TAA can effectively modulate the crystallization kinetics, with TAA demonstrating the most pronounced retarding effect. The enhanced retarding effect with TAA can be attributed to its stronger complexation with lead within the perovskite. From the density functional theory (DFT) calculations, the interaction energy between TU and PbI₂ was calculated as -1.230 eV , while for TAA, it measured -1.462 eV .

Another example is the work by Zheng et al.,¹¹⁴ who focused on the influence of thiourea (TSC) and DMSO on (BA)₂(MA)₃Pb₄I₁₃ 2D perovskites. These additives were employed with the dual purpose of controlling the crystallization process and enhancing the vertical orientation of grains, thereby improving the performance of 2D perovskite solar cells. The underlying hypothesis was that if the crystallization process yields a multitude of crystallites, then it results in a perovskite with a lower degree of grain orientation. Conversely, if the process generates fewer but larger crystallites, then it leads to a highly oriented perovskite with larger grains. Their initial analysis, performed via DLS, revealed that the combined use of TSC and DMSO causes an increase in colloidal size compared to the individual application of these additives. Moreover, examination of cross-sectional SEM images unveiled that the most favorable morphology was achieved when TSC and DMSO were employed together, as it exhibited minimal grain boundaries. To analyze the degree of orientation in these materials, the authors performed GIWAXS on films both with and without additives, as depicted in Figure 13D. In the absence of additives, scattering rings were observed, indicating random grain orientation. Conversely, in the presence of both TSC and DMSO, these rings were notably absent, and only Bragg spots were discernible, with pronounced intensity in the out-of-plane direction. The authors stated that the two additives assumed distinct roles: TSC, functioning as a robust Lewis S-donor, exhibited the capability to complex strongly with the perovskite precursors and also orient them, whereas DMSO, acting as a Lewis O-donor, played a pivotal role as an aggregation retardant.

Rao et al. investigated the impact of hexamethylphosphoramide (HMPA) on tin-based perovskites.¹¹⁵ This molecule, being a strong Lewis base and exhibiting a high vapor pressure, is able to coordinate with tin during the crystallization process. DFT calculations revealed a notably higher interaction energy between HMPA and tin compared to other solvents, such as DMSO, signifying enhanced structural stability. As depicted in Figure 14A, the pathway involving the formation of the SnI₂-HMPA complex is favored, as evidenced by the higher modulus of the binding energy. The authors validated the positive influence of this additive by employing GIXRD measurements during spin coating on solid films prepared with and without HMPA (Figure 14B). They noted that the presence of HMPA results in a reduced intensity of the peak at 14° compared to the additive-free scenario, indicative of a deceleration in the crystallization process. As a result, it can be seen from UV-vis and PL analysis in

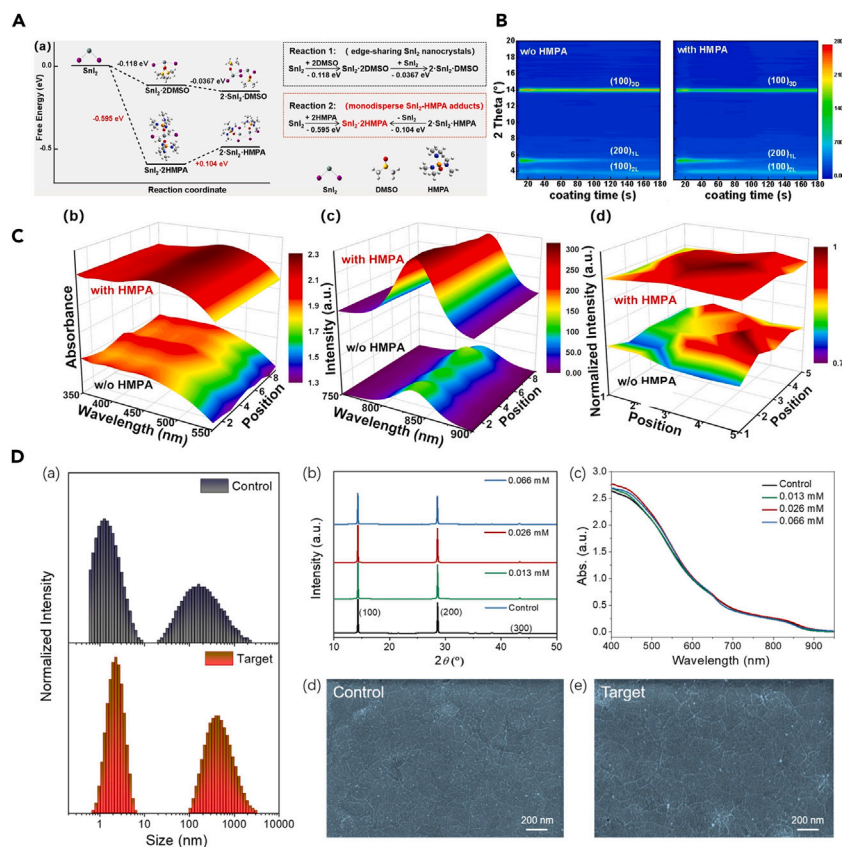


Figure 14. Study of the effect of CSs on tin-based perovskites

(A) DFT free energy as a function of the reaction coordinates for SnI_2 -DMSO and SnI_2 -HMPA complexes (adapted with permission from Rao et al.¹¹⁵).
 (B) *In situ* GIXRD of perovskite films with and without HMPA (adapted with permission from Rao et al.¹¹⁵).
 (C) 2D UV-vis (b) and PL (c and d) of the crystallized films with and without HMPA (adapted with permission from Rao et al.¹¹⁵).
 (D) DLS analysis of the perovskite precursor suspensions with and without TU (a), XRD patterns with different concentrations of TU (b), absorbance spectra with different concentrations of TU (c), and surface SEM images of the perovskite films without (d) and with (e) TU (reprinted from Wang et al.¹¹⁶ Copyright 2022, with permission from Elsevier).

Figure 14C that films prepared with HMPA exhibited higher and more uniform absorption and PL than their counterparts prepared without the additive.

Wang et al. implemented TU molecules into the MASnI_3 perovskite suspension to serve as regulators of crystallization kinetics.¹¹⁶ DLS analysis in Figure 14D (a) reveals a notable rightward shift in both peaks compared to the control sample, signifying changes in particle size distribution. Furthermore, XRD analysis in Figure 14D (b) demonstrates heightened peak intensities related to the (100), (200), and (300) crystallographic planes, along with a reduced FWHM, indicative of enhanced crystallinity. This is true up to a concentration of 0.026 mM. Concurrently, it is apparent that the absorbance in Figure 14D (c) follows a similar trend, with the peak absorbance occurring at a concentration of 0.026 mM. The SEM image of the sample containing the additive in Figure 14D (e) exhibits larger and coarser grains compared to the pristine sample (Figure 14D, panel d), providing evidence that the presence of the additive results in decelerated crystallization, subsequently leading to the enlargement of

grain sizes. Liu et al.¹¹⁷ explored the utilization of melamine with a very similar strategy. They observed that melamine, characterized by the presence of three $C\equiv N$ and NH_2 functional groups, possesses the capability to engage in acid-base interactions and hydrogen bonding with Sn^{2+} ions. The introduction of melamine induces the formation of complexes that significantly impact the growth dynamics of the perovskite material. DLS analysis revealed an increase in the average colloidal size from 176 to 376 nm with the addition of melamine. Correspondingly, XRD analysis demonstrated heightened peaks and reduced FWHM upon the inclusion of melamine. This effect correlated with an observed increase in grain size, as evidenced by SEM images.

CHALLENGES AND FUTURE DIRECTIONS

In this comprehensive review, we elucidated the intricate interplay between the colloidal properties of perovskite dispersions and the underlying mechanisms governing solid film formation. In particular, we showed how additives may be classified into three types, namely, CRIs, CIMs, and CSs, based on their effects on colloidal suspensions.

For future applications, a potential challenge might be the use of numerical simulations as a predictive tool for the aggregation and crystallization dynamics of perovskites from a colloidal suspension. Multiple simulation techniques can be employed, each with distinct advantages. For instance, molecular dynamics can solve the classical equations of motion for all molecules or particles in the system without simplifying assumptions.^{118,119} This method excels in accurately describing time-averaged equilibrium properties, such as diffusion coefficients and reaction kinetics. Brownian dynamics (BD) is a numerical technique where the solvent molecules are replaced by stochastic forces. As opposed to molecular dynamics, it proves to be extremely powerful when studying systems with diverse timescales, such as suspensions with particles of significantly different sizes.^{120–122} The Monte Carlo method is a stochastic technique that is especially useful for studying equilibrium properties of a system.^{123–125} This method relies on Markov chain theory, which asserts that the transition of a system from one state S_1 to another state S_2 is solely contingent upon the current state S_1 and is independent of any preceding states. Stochastic rotation dynamics is a mesoscopic simulation technique that allows translational and rotational motion (streaming and collision steps) of particles with specific masses.¹²⁶ This method exhibits considerable versatility and allows for the incorporation of hydrodynamic interactions among colloids, which holds significant relevance for the study of colloidal suspensions. Finally, the lattice Boltzmann method divides the space into discrete lattices where particles execute movement and collisions.¹²⁷ This algorithm is reminiscent of stochastic rotation dynamics but also offers the possibility of being parallelized.

Due to the inherent complexity of perovskite systems, employing BD offers a viable approach, given its capacity to mitigate solvent effects by introducing stochastic terms into the balance equations. Another valuable approach that can be coupled with BD is the coarse-grained method, which aims to create a simpler representation of complicated systems while keeping the fundamental chemical/physical features. This method involves aggregating atoms into entities called beads, which interact based on specific interaction potentials. Beads can vary significantly in composition, ranging from a few atoms to complex structures like proteins¹²⁸ or polymer chains.¹²⁹ This approach offers versatility and simplifies simulations by reducing the system's component complexity. In the context of perovskite systems, beads could represent nanocrystalline perovskite colloids, as schematized in [exploring the nature of perovskite colloids](#), and the suspension of perovskite precursors may be construed as a suspension of such beads.

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AUTHOR CONTRIBUTIONS

D.A. prepped the first draft of the manuscript. All the authors revised and provided input to the manuscript preparation.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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